

Why Does an IPO Affect Rival Firms?

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IPO firms' rivals tend to experience performance declines following an IPO in the industry. Why? We estimate a dynamic structural oligopoly model to distinguish between alternative theories that can explain an industry's evolution post-IPO. We find that most changes in rivals' performance are due to industry trends that also drive IPOs. However, we also find some "competitive" IPOs where the IPO enhances the IPO firm's performance at the expense of competitors. These findings help reconcile prior evidence of average performance reductions of both IPO firms and their rivals with well-known cases in which firms have benefited from going public. (*JEL* G30, G32)

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Most research on initial public offerings (IPOs) has focused on the short- and long-term performance of IPO firms, but there are striking patterns in the post-IPO performance of rival firms as well. On average, entire industries see their values and profitability decline following the typical IPO. For example, Hsu, Reed, and Rocholl (2010; HRR) report that public incumbents experience a 3.3% decline in sales growth and a 2.9% reduction in profitability during each of the 3 years following an IPO in the industry. Competitors' also exhibit negative abnormal stock returns near completed IPOs.

Why does performance decline among an IPO firm's competitors? The vast IPO literature offers many theories describing when the benefits from going public exceed its costs, which might help explain rivals' post-IPO performance. Three such possibilities are (1) a firm may undertake an IPO to lower its financing costs, making it a stronger competitor; (2) founders and may wish to retire or sell off their holdings; and (3) an IPO may occur when there is a shock to industry fundamentals, making the benefits of remaining private less attractive.

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Taking a “one-size-fits-all” approach to explaining the patterns we observe can limit our understanding of why changes occur in industry performance following IPOs. Rather than try to determine which theory is “right,” we start with the assumption that each one might hold for at least some IPOs. We then conduct an event-by-event analysis of the evolution of the IPO firm’s industry pre-and post-IPO to determine which theory best fits the data for each individual IPO. The results indicate that industry-wide changes explain rivals’ post-IPO performance in the majority of cases. However, in 5%–10% of cases, a stronger IPO firm with new access to cheaper public financing is the most likely driver.

Understanding which theory best explains the data requires having some idea as to how individual firms would have fared absent the IPO. In this paper, we use a dynamic model of industry competition based on Spiegel and Tookes (2013a) to provide a baseline. The model takes place in continuous time and generates quarter-by-quarter performance forecasts for each firm in the industry. We map differences between these forecasts and actual performance for a particular firm or industry to hypotheses capable of explaining the causes and consequences of IPOs. We then produce a frequency distribution of the applicability of each theory across events.

In our application, the model creates parameter forecasts that are consistent with a null hypothesis that all firms in an industry change over time at the same rate, in proportion to their pre-IPO values. That is, if the profitability parameter a_i for firm i transitions from a_i to ka_i at a rate ψ , then all firms in the industry must see the same transition; a_j to ka_j at the rate ψ . Importantly, if the IPO firm’s fundamentals are transitioning in a way that is different from its rivals, we can reject this null hypothesis via forecast error tests. If the observed decline in rivals’ performance post-IPO is due to the increased strength of the IPO firm, then forecasts restricted to treating all firms in the industry (including the IPO firm) in the same manner should yield biased forecasts with errors that increase over time. Further, the bias for the IPO firm should be in a direction opposite its rivals. Our analysis indicates this is not the case. On average, forecast errors for the IPO firm are no different from those of its industry peers. This supports the hypothesis that IPOs foreshadow future industry changes rather than cause them.¹ Our use of forecast errors from a structural model to shed light on the drivers of industry performance dynamics is novel and provides insights that would be impossible to obtain using standard methods.

Negative industry-wide trends appear to be the dominant explanation for the patterns that we observe, but there is also substantial variation across IPO events. We can use the estimated parameters and differences in the forecast errors between rivals and the IPO firm to learn more about cross-sectional variability

¹ The finding that post-IPO performance of newly public firms is no different from industry peers does not imply that going public was a bad decision. An IPO may have been the firm’s best policy. The question addressed here is separate and focuses on whether the primary driver behind the decline in rival performance is due to a change in the industry’s overall environment or due to the newly public firm’s increased competitive strength.

in the likely causes and consequences of IPOs. For example, suppose the effects that we observe are due to a firm becoming a more formidable competitor by going public. In this case, IPO firms should perform better than expected while its rivals will perform worse. Using the 5-year estimation horizon as an example, we find evidence consistent with this idea in approximately 8% of the events that we analyze. By comparison, in 54% of all events, the evidence is consistent with an overall negative industry trends that triggers the IPO. Although IPOs with competitive effects are a clear minority, the ability to use the model to identify them highlights an important advantage of the empirical approach that we use. One way to interpret the data is to ask how a Bayesian would update her priors given the empirical distributions that we report. Our findings point toward posterior distributions with substantial weight on the industry trend hypothesis.

Our analysis offers some novel insights into why firms may decide to go public and why the average industry subsequently experiences declining overall profits. For example, a driving force behind the model is consumer loyalty, or the speed at which customers are willing to switch between firms' products. Using data on the evolution of market shares within the industry and across time, we estimate customer loyalty, and we find that it declines substantially following IPOs.² We hypothesize that increased product commoditization reduces the benefits of maintaining private status, causing firms to go public earlier in their lives. We take this prediction to the data by relating the estimated change in consumer loyalty near IPO events to the size at which firms go public. Within industries, we find that firms go public with smaller market shares when the post-IPO decline in consumer loyalty is bigger. In the median industry, the increased commoditization following an IPO reduces the estimated market share at which the next firm in the industry goes public by about 15%. This supports the idea that the IPO event is a response to anticipated industry changes, not their cause. It also helps reconcile evidence from HRR, Chemmanur and He (2011; CH), and Chod and Lyandres (2011; CL) that IPO firm rival performance declines with the earlier literature showing that is true of the IPO firm as well (Jain and Kini 1994; Teoh, Welch, and Wong 1998; Ritter and Welch 2002). That is, both the newly public firm and its rivals see a deterioration in their performance post-IPO.

While our evidence points to industry-wide developments as the main culprit behind rival firms' post-IPO performance declines, it is still possible that the IPO firm caused the industry-wide changes. We consider the possibility that, by not going public, the IPO firm would have experienced an even worse fate. In that case, the IPO firm might find it optimal to trigger the industry's decline by

² The packaged iced tea industry is an example of this type of market evolution. Around the time of Snapple's IPO, several firms entered with their own teas, leading consumers to switch more readily between corporate offerings. Backman (2015) provides a brief history. Burns (1996) discusses how rival drink makers (like PepsiCo and Coca-Cola) entered the iced tea industry with their own iced tea versions.

going public. The model lets us determine the plausibility of the “IPO trigger” hypothesis given the magnitude of the gains and losses involved. In particular, would incumbent rivals be better off purchasing still private competitor to avoid triggering the post-IPO performance decline? We find that the magnitudes of the incumbents’ post-IPO losses are so large that in nearly all cases the largest firm in the industry would be better off buying the IPO firm to prevent the decline, even if that entailed doing so at a considerable loss. Moreover, in many cases, it would even pay to buy the IPO firm and then shut it down. Overall, the IPO trigger hypothesis seemingly requires nearly dysfunctional private equity and takeover markets.

Some of the most important insights in this paper would be difficult to obtain using existing models. For example, many dynamic structural models evolve from the real options literature. Notable papers include those of Aguerrevere (2003, 2009), Novy-Marx (2007), and Carlson et al. (2014). In these dynamic models, firms compete via Cournot competition by expanding their production capacity of the market’s single homogenous good. Increasing capacity lets a firm sell more of its product but also reduces the overall price. Each paper in this area then considers a particular setting (competitive, symmetric oligopoly, etc.) to analyze. In principle, it is possible to redeploy one of these models to examine the going public decision, at least in cases where the IPO is used to expand the firm’s operations. However, we think the model used in this paper offers a more flexible setting for understanding IPOs. For example, most industries have firms that compete via the production of heterogenous goods and operate with varying degrees of efficiency. The real options models impose a significant degree of symmetry among the competitors that would make adapting them to such setting rather challenging. In contrast, our approach allows both heterogenous goods and a multiplicity of operating efficiencies.

How well does this paper’s empirical model capture industry dynamics? The appeal of the Spiegel and Tookes (2013a) framework is that it captures firm heterogeneity in an n -firm industry and is easy to estimate; however, our findings and conclusions are relevant only if the model captures meaningful variation in observed changes to corporate profits and values before and after a rival’s IPO. We conduct out-of-sample tests of model performance and find strong support for the model. Period-ahead forecast tests indicate that, if you have to restrict yourself to one or two independent variables, then the structural model’s predicted value and profit changes should always be among them. Most of the time, if you are restricted to just one, it would be the structural model’s forecast. These out-of-sample tests also provide a check against results that are driven by misspecification bias. Misspecified and over fit models typically yield predictions about the future that are little more than white noise. (See, e.g., the Goyal and Welch (2008) critique of the equity premium literature.) The out-of-sample tests show that our model better predicts future changes in the industry than do model-free alternatives like AR processes as well as those that employ variables seen in other papers in the IPO literature.

Additionally, Harvey and Collier (1977) misspecification tests of the model on the full Compustat database also indicate that the model forecasts are not just noise. Overall, the model's failure rate equals the size of the test plus approximately 1%. If the model were perfect, the failure rate should equal the test size. The main body of the paper discusses this in further detail.

Our forecast error tests can be thought of as a generalized differences-in-differences (DD) test. In a DD analysis, one compares the difference in some value between dates 0 and 1 across two groups. In essence, the model takes the date 0 value as a "forecast" of the date 1 value. Conclusions are then drawn based on these forecast errors across groups. In this study, the forecast at date 1 is not the date 0 value, but the value predicted by the dynamic model. The errors are the difference between the model's forecast and the realized date 1 value.

1. Model

We adapt Spiegel and Tookes (2013a) to the IPO setting. The basic setup and intuition are outlined below, with details in the appendix. The main goal is to produce an estimable structural model that can be used to forecast industry performance variables in the IPO event setting.

Consider an industry containing n firms that produce a heterogeneous product. Firm i competes with its rivals for market share (m_i) by spending funds $u_i(t)$ at time t on customer acquisition. Advertising is perhaps the most obvious way firms acquire market share, but $u_i(t)$ might also include research and development to improve product design, as well as capital expenditures that create outlets closer to where customers shop. The Spiegel and Tookes (2013a) model assumes that market share evolves over time via:³

$$dm_i(t) = \phi \left[\frac{s_i u_i(t)}{\sum_{j=1}^n s_j u_j(t)} - m_i(t) \right] dt. \quad (1)$$

The variable ϕ also can be thought of as consumers' perception of product heterogeneity. Low values of ϕ imply high customer loyalty (and high product heterogeneity). The variable s_i captures a firm's customer acquisition effectiveness. Higher values of s_i imply that a firm achieves a greater "bang for its buck" when trying to gain market share via spending $u_i(t)$. Intuitively, Equation (1) says that relative spending, determined by spending amount and effectiveness, drives variation in market shares. Larger firms have more to lose,

³ At the limit, where every firm spends zero on customer acquisition, the ratio becomes undefined. One can address this by assuming that firms are endowed with a small free level of customer acquisition, which is equivalent to assuming a firm's product has a subset of very loyal customers. Ultimately, one can show that adding this to the model does not materially alter any of its results.

and smaller ones have more to gain. Note that a variant of this law of motion is standard in the empirical marketing literature (for a review, see Dockner et al. 2000). The main differences in our version are that we allow for N firms (rather than just a duopoly) and we also add a denominator that represents total effective spending in the industry, making the law of motion unit free.

Let ς represent the measure of an industry's size measure (exogenous)⁴ and assume it follows the law of motion

$$d\varsigma = gdt + \sigma dw, \quad (2)$$

where dw is a standard Brownian motion.⁵ In the model, instantaneous corporate profits are proportional to industry size and are given by

$$e^{\varsigma} \pi_i(t) = e^{\varsigma} [\alpha_i m_i(t) - u_i(t) - f_i]. \quad (3)$$

The parameter α_i translates a unit of market share into corporate profits gross of its spending on customer acquisition and its fixed operating costs f_i . Each firm seeks to maximize its expected present discounted value $V_i(m, \varsigma, t) = E \left[\int_{t=0}^{\infty} e^{\varsigma - rt} \pi_i(t) dt \right]$, where r (assumed to be greater than g) is the discount rate. In principle V_i can be a function of the vector of market shares, m . However, we will guess that it is a function of m_i and then confirm that this guess is a solution to the value function. Appendix Section A.1 shows that a change in the probability measure lets one further simplify the problem by writing the value function as $V_i(m_i, t)$, which drops the ς term and with it the second order terms in the HJB equation that determines the solution to the optimal control problem.

The empirical work will compare the model's forecasted valuation changes (V) to observed changes. We emphasize that V refers to a firm's enterprise value. An investor in the stock expects to earn the risk-adjusted market return, which should follow a random walk. However, this need not hold for a firm's enterprise value. In the model, the firm pays out its profits in excess of its investment as a dividend or takes in capital when expenses are less than costs. The resultant expected return to an investor will indeed equal the discount rate. However, the firm's enterprise value may grow at a higher or slower rate, depending on overall industry performance and the firm's change in market share.

Assume that, prior to the IPO, firms in the industry are endowed with a set of parameter values for their profits per unit of market share $\alpha^* = (\alpha_1^*, \dots, \alpha_n^*)$,

⁴ In the model, corporate spending on customer acquisition does not influence the overall industry's growth rate. Because of this assumption, the model better describes a mature oligopoly than a new and rapidly growing one. For example, if a company develops a better refrigerator, it is unlikely that many households will then add an additional refrigerator to their house. Instead, sales are more likely to come at the expense of rivals. Many industries function like this. Adding an interaction between the u_i and ς is clearly possible, but doing so comes at the cost of a closed-form solution.

⁵ Note that Spiegel and Tookes (2013a) do not consider stochastic industry size. We include it here for a richer comparison with the previous IPO literature.

spending efficiency $s^* = (s_1^*, \dots, s_n^*)$, and fixed operating costs $f^* = (f_1^*, \dots, f_n^*)$. For simplicity, assume that firm n conducts the IPO. Over time, the industry will adjust to the new post-IPO parameter sets $\alpha = (\alpha_1, \dots, \alpha_n)$, $s = (s_1, \dots, s_n)$, and $f = (f_1, \dots, f_n)$. These new parameters may reflect changes in the overall industry structure. They could also reflect the competitive effects of the new, stronger, publicly financed IPO firm. Alternatively, they might reflect the competitive effects of rivals' new ability to observe some of the IPO firm's once-private information. The model does not take a stand regarding the industry's progression following the IPO. Parameters may evolve in ways favorable or harmful to firms. The overall impact on firm and industry value will depend on not only the changes occurring in the industry but consumer demand and the competitive responses of all firms in the industry.

Without imposing any restrictions as to whether an IPO triggers an industry's evolution or whether it is good or bad for the industry, assume that the transition to the new parameter set occurs at an exponential rate ψ . Consequently, at time t , a firm's actual profits per unit of market share, spending efficiency, and fixed costs are given by $\tilde{\alpha}$, $\tilde{s}$, and $\tilde{f}$:

$$\tilde{\alpha} = \alpha + e^{-\psi t} \Delta\alpha, \quad (4)$$

$$\tilde{s} = s + e^{-\psi t} \Delta s, \quad (5)$$

and

$$\tilde{f} = f + e^{-\psi t} \Delta f, \quad (6)$$

where $\Delta\alpha = \alpha^* - \alpha$, $\Delta s = s^* - s$, and $\Delta f = f^* - f$. This lets the model incorporate all three elements into the $-\delta V_i$ term in the HJB equation. To reduce notational clutter further, let $\delta = r - g - \frac{1}{2}\sigma^2$.

Write the problem of finding the optimal amount to spend on consumer acquisition as

$$0 = \max_{u_i} (\alpha_i + e^{-\psi t} \Delta\alpha_i) m_i - f_i - e^{-\psi t} \Delta f_i - u_i + V_i^m \phi \left[\frac{\tilde{s}_i u_i}{\sum_{j=1}^n \tilde{s}_j u_j} - m_i \right] - \delta V_i + V_i^t. \quad (7)$$

In Equation (7), superscripts on the value function V indicate partial derivatives with respect to market share (m_i) and time (t). Like most papers that employ continuous time game theoretic models, the above setup seeks to find a Markov equilibrium. There may be, and likely are, other non-Markovian equilibria.⁶

To solve (7) assume V_i has the form

$$V_i(m_i, t) = a_i(t) + b_i(t) m_i(t). \quad (8)$$

⁶ See Dockner et al. (2000) for a discussion of Markov and non-Markov equilibria in continuous-time game theoretic models.

⁷ Because firms are risk neutral and the value function is linear in the state variable (m), a random term can be added to the market share's law of motion (Equation (1)) without changing the model's solution. See Dockner et al. (2000) for this result.

The empirical work that follows is based on the following equations. Define

$$\hat{\alpha}_i = (\phi + \delta + \psi)\alpha_i + (\phi + \delta)\Delta\alpha_i e^{-\psi t} \quad (9)$$

and

$$\hat{z} = \sum_{j=1}^n \frac{1}{\hat{\alpha}_j \tilde{s}_j}. \quad (10)$$

Then the solution to u_i is

$$u_i = \frac{\hat{\alpha}_i \phi (n-1) [\hat{\alpha}_i \tilde{s}_i \hat{z} - (n-1)]}{(\phi + \delta + \psi)(\phi + \delta) (\hat{\alpha}_i \tilde{s}_i \hat{z})^2}. \quad (11)$$

The solution for b_i is

$$b_i = \frac{\alpha_i}{\phi + \delta} + \frac{\Delta\alpha_i e^{-\psi t}}{\phi + \delta + \psi}. \quad (12)$$

Finally, the solution to a_i comes in the form of nonlinear ordinary differential equation (ODE). While it does not in general admit a closed-form solution, it does so if one assumes that $\hat{\alpha}_i = ((\phi + \delta + \psi)\alpha_i + (\phi + \delta)e^{-\psi t}\Delta\alpha_i) = \alpha_i((\phi + \delta + \psi) + (\phi + \delta)k_\alpha e^{-\psi t})$ and $\tilde{s}_i = s_i(1 + k_s e^{-\psi t})$ for industry-wide constants k_α and k_s . These constraints imply that each firm experiences a proportional change to its profits per unit of market share and marketing capabilities post-IPO. In this case, one can show that the $\hat{\alpha}_i \tilde{s}_i \hat{z}$ terms reduce to the constant $\alpha_i s_i z$, where z is defined as $\sum_j (\alpha_j s_j)^{-1}$. This then leads to the closed-form solution for the a_i of

$$a_i = -\frac{f_i}{\delta} + \frac{\alpha_i \phi [\alpha_i s_i z - (n-1)]^2}{\delta(\phi + \delta)(\alpha_i s_i z)^2} + e^{-\psi t} \left[\frac{\alpha_i k_\alpha \phi [\alpha_i s_i z - (n-1)]^2}{(\delta + \psi)(\phi + \delta + \psi)(\alpha_i s_i z)^2} - \frac{\Delta f_i}{\delta + \psi} \right]. \quad (13)$$

For empirical purposes, all that is necessary is the above solution. It allows us to estimate model parameters and to create forecasts of firm value (from Equation (8)) and profitability over time.

In interpreting the solution, it is useful to point out that, while it is true that each firm in the industry experiences the same shock that results in a proportional change in its profits per unit market share, they will respond differently to the shock because they are heterogeneous. Equation (11) provides an example: increases in α_i increase equilibrium spending for large firms but decrease it for very small ones. The model fully incorporates this heterogeneity. In empirical analyses, this means that finding a closely matched control firm within the industry is unnecessary. One can use the same model to compare the dynamics of very different firms.

⁸ See A Step-by-Step Guide to Solving the Model in Spiegel and Tookes (2013b) for the derivation of Equations (9) through (12), when there is no exponential drift component. Adding in and then solving for the exponential drift is then straightforward.

1.1 A comparison with the real options literature

Before proceeding, it is useful to discuss our choice of the Spiegel and Tookes (2013a) framework, in light of the related literature on real options in imperfectly competitive product market settings. The models in the real options literature focus on Cournot homogenous goods markets in which firms decide whether to expand production. The firms either have identical production and expansion technologies or they compete in two-firm industries. For firms, the relevant trade-off is that expansion lets a firm sell more but also reduces the price of the good being sold and increases the cost of the capital that is employed. The expansion decision depends on the competitive setting the model employs.

Aguerrevere (2003, 2009) consider putty-clay capacity choice problems in a market with linear (Aguerrevere 2003) and nonlinear (Aguerrevere 2009) demand. The analysis initially examines a monopolist's behavior and then considers an industry with n identical firms. Novy-Marx (2007) replaces the set of identical n firms with a continuum. These firms produce an amount equal to their current capacity and compete by altering that capacity. Like Aguerrevere's models, firms again share an identical production and expansion technology. Novy-Marx (2007) does create a level of asymmetry by allowing for variation in initial firm capacities. (However, if all the firms were to start at an identical size, they would remain identical through time.) Carlson et al. (2014) consider a Cournot duopoly. They then consider three settings in which (1) one firm can expand and contract but the other cannot, (2) both firms have a single expansion option, and (3) where they have a single contraction option. Asymmetry arises from the relative cost each firm faces to alter its capacity level and in their costs of production. Their primary focus is on how these options affects each firm's overall risk.

Similarities between the models in the real options literature and the Spiegel and Tookes (2013a) framework include the focus on Markov perfect equilibria, stochastic growth at the industry level and size of the strategy space. All of these continuous time models restrict firms to employing a single control variable. In the real options literature, the control variable of interest is capacity. In this paper, the control variable of interest is spending to increase market share (this could include advertising, R&D, or product design, but not capacity, which is unconstrained at market prices).⁹ In addition, and as described in the Online Appendix, the industry's one private firm has potentially two controls if going public is endogenous.¹⁰

⁹ In the IPO setting, the primary motivations for going public include reducing the cost of capital, establishing a market price, cashing out, and timing overall market conditions (see, e.g., Pagano, Panetta, and Zingales 1998; Brau and Fawcett 2006). Based on the existing literature, increasing investment in capacity is seemingly less relevant.

¹⁰ The model that we use in this paper begins at the IPO date, taking the decision to go public as given. Leading up to that date, the firm likely had to consider both changes in its own status and those affecting the whole industry. In the Online Appendix, we consider a version of the model that endogenizes the IPO decision. Doing so allows us to characterize the trade-offs that private firms make and to consider how these might affect how we interpret the results from estimating the main model.

The most important difference between the model used in this paper and those in the real options literature is that the Spiegel and Tookes (2013a) setting allows for firm heterogeneity in an industry with n firms. This provides clear advantages when we take the model to the data. In reality, corporate profits vary across firms along a number of dimensions, including their marginal and fixed costs of production and their spending effectiveness.¹¹

1.2 Model forecast errors

The empirical tests described in the next section use a combination of the model's estimated parameters and its forecast errors. Forecast errors compare the model's predicted evolution of the industry post-IPO with what actually happens. These differences allow us to see the degree to which an event, in this case an IPO, pushes the firms in an industry off the path one might have otherwise predicted. As such, they are a way to characterize an event's influence.

To obtain the forecast errors, we begin by estimating the model parameters using data from the IPO firm's rivals during the quarters surrounding the IPO event. Based on these estimates, the model generates predictions regarding future performance of both the IPO firm and its rivals. The difference between these projections and each firm's actual performance are the forecast errors. Depending on the IPO firm's reason for going public, certain systematic differences in these errors should develop.

2. Why Go Public? The Ex Post Industry Implications

Researchers have proposed several hypotheses to explain why a private firm may go public. A reduced form analysis that aggregates data across industries and IPOs may find support a given theory; it will likely have trouble distinguishing which theory fits which event the best or how frequently one theory or another applies. We use the framework in this paper to test the validity of five candidate hypotheses.¹² Along with each hypothesis, we provide specific empirical implications for how the industry data should evolve post-IPO. The discussion below assumes that the IPO can: reduce financing costs via an increase in α_{IPO} , the profitability of the IPO firm; increase fixed costs for the IPO firm (increase in f_{IPO}) due to increases in regulatory costs for public firms; and reduce the effectiveness of spending (s_{IPO}) due to the inability to keep competitive secrets private post-IPO.

¹¹ Another difference is the solution's interpretation. As noted by Back and Paulsen (2009), articles that follow along the lines of Grenadier (2002) solve for a Markov perfect open-loop Nash equilibrium. In the literature discussed here, both Aguerrevere (2003, 2009) papers fall into this category. Unlike the closed-loop equilibrium examined in Carlson et al. (2014) and in this paper, open-loop equilibria are not subgame perfect.

¹² The list, although not exhaustive, covers the most likely candidate explanations.

2.1 Hypotheses and implications

H0 (Null): Reasons outside of the model.

IPOs occur when a firm reaches a critical scale, or for noneconomic reasons that are outside of our model (e.g., a founder's retirement). Implications:

- (1) Model parameters: If the IPO occurs when a firm reaches critical scale or for noneconomic reasons, then we do not expect significant changes in industry profitability or consumer responsiveness following the IPO (i.e., no significant changes in parameter value ϕ and k not significantly different from 1).
- (2) Forecast errors: Under *H0*, the forecast errors for the IPO firm are either negative or similar to those of its rivals. If the IPO is occurring at the optimal time, then the firm becomes large enough that the value of lower public financing costs (increase in α_{IPO} , the profitability of the IPO firm) exceeds the value from being able to keep financials private (reduced value of s_{IPO}) and the fixed cost of going public (increase in f_{IPO}). In this case, the value and profit forecast errors should be similar for the IPO firm and its rivals. If the IPO is occurring for noneconomic reasons, the IPO firm's forecast errors should be negative (i.e., we expect to observe underperformance because the IPO is not occurring at the optimal time).¹³

H1: IPOs as Causal Events – Firms go public to increase their competitive strength.

If IPOs make newly public firms stronger, increasing α_{IPO} to a degree that the gains from going public exceed the costs imposed via a lower s_{IPO} and increased f_{IPO} . The increase in the IPO firm's competitive strength leads to a decline in rival values. Implications:

- (1) Model parameters: Industry conditions are deteriorating for rivals. This implies a significant $k < 1$ or significant $\phi - \phi_0 > 0$.
- (2) Forecast Errors: Value and profit forecast errors for the IPO firm should be positive and significant (i.e., the IPO firm should do better than expected). Value and profit forecast errors for rival firms should not be significantly positive.

H2: IPOs as canaries: Industry-wide shift in consumer preferences triggers a firm to go public.

While an IPO raises α_{IPO} , lowers s_{IPO} , and increases f_{IPO} , the primary impact on rival firms comes via increased product homogenization over time; an increasing value of ϕ . Implications:

¹³ The Online Appendix models this trade-off explicitly.

- (1) Model parameters: The estimated change in the consumer responsiveness parameter ($\phi - \phi_0$) is positive and significant, and k is not significantly greater than 1.
- (2) Forecast errors: The forecast errors for the IPO firm and industry are not significantly positive, nor are they significantly different from one another.

H3: IPOs as Canaries – Deterioration in industry-wide profitability triggers a firm to go public.

While an IPO raises α_{IPO} , lowers s_{IPO} , and increases f_{IPO} , the primary impact on rival firms comes via changing production technologies that decrease every firm's α_{IPO} over time. Implications:

- (1) Model parameters: The value of k is significantly > 1 and $\phi - \phi_0$ is not significantly negative.
- (2) Forecast errors: The forecast errors for the IPO firm and rivals are not significantly positive, nor are they significantly different from one another.

Note that *H2* and *H3* are not mutually exclusive. Changes in consumer loyalty can coincide with (or even cause) changes in profitability in the industry. For example, firms might offer discounts if consumers begin to view firms' products as more homogenous.

H4: IPOs as Canaries 2 – Improving industry conditions triggers a firm to go public

While an IPO raises α_{IPO} , lowers s_{IPO} , and increases f_{IPO} , the primary impact on rival firms comes either via changing production technologies that increase every firm's α_{IPO} over time or from decreased product homogenization over time, a declining value of ϕ . Implications:

- (1) Model parameters: For the estimated parameters either the value of (a) $\phi - \phi_0$ is negative and significant and k is not significantly less than 1 or (b) k is significantly > 1 , and $\phi - \phi_0$ is not positive and significant.
- (2) Forecast errors: The forecast errors for the IPO firm and its rivals should not be significantly negative nor should they be significantly different from one another.

2.1.1 Ambiguous changes. While the empirical model considers either a positive or a negative shock to profitability or consumer demand near IPO events, it is possible that industry parameters are changing in ways that both benefit and hurt firms. This implies one of the following: (a) $\phi - \phi_0$ is negative and significant and k is significantly less than 1; (b) k is significantly greater than 1, and $\phi - \phi_0$ is positive and significant; (c) industry fundamentals are

deteriorating (improving) based on parameter estimates, but forecast errors indicate that firms are performing better (worse) than predicted; or (d) no change in industry fundamentals, but the IPO firm is performing differently from peers (i.e., is a negative correlation in the forecast errors of the IPO firm and the industry) or there is insufficient data to conduct forecast error tests. In categorizing IPO events, we place IPO events in which industry fundamentals are changing in ambiguous ways in the “outside of the model category” ($H0$).

2.2 Mapping the implications to empirical tests

In the next section, we implement empirical tests of the hypotheses described above. In the tests, we require unambiguous and statistically significant changes in model parameters in order to classify events into those that correspond with deteriorating and improved industry conditions (Hypotheses 1 through 4). Deteriorating industry conditions imply declining profitability ($k < 1$) and/or a post-IPO increase in consumer responsiveness ($\phi - \phi_0 > 0$). The statistical significance requirement is met if at least one of these parameter estimates are significant at the 10% level.¹⁴ Bayesian updating tests are also conducted which obviate the need to select a significance level.

The unambiguous deterioration condition is satisfied if we also do not observe a significant change in one of the parameter estimates that suggests improving industry conditions (e.g., settings where k is significantly less than 1 but $\phi - \phi_0$ is also significantly less than 0). Similarly, improving industry conditions imply an increase in industry profitability ($k > 1$) and/or a decrease in consumer responsiveness ($\phi - \phi_0 < 0$). All events in which we do not observe significant changes in model parameters or in which there is ambiguity in the changes that we observe are classified as “outside of the model” events ($H0$).

In the forecast error tests, we examine the difference between actual and model-implied value and profitability changes. In classifying forecast errors into those that are positive or negative (i.e., over- and underperformance, respectively), we require both statistical significance and no ambiguity. For example, forecast errors are defined as positive if value (profitability) change forecast errors are significantly positive at the 10% level and profitability (value) change forecast errors are not significantly negative. Forecast errors of IPO firms and their rivals are classified as similar to one another if we do not observe a negative and statistically significant correlation between them.

3. Empirical Analysis: Competitive Effect of an IPO

An important advantage of the model is that it characterizes the value dynamics resulting from changes in the competitive structure of the industry in a way that

¹⁴ The Online Appendix shows results of analyses in which we define statistical significance based on 5% and 1% levels rather than the 10% level. The main findings are not driven by the choice of significance threshold.

is amenable to empirical estimation and testing. This section presents estimates of the model's key parameters.

3.1 Profitability transition

The profitability equation provides a structure for estimating firm-specific and industry-wide parameter values. Recall that the basic profit function is given by: $\pi_i(t) = e^{\varsigma(t)}(\alpha_i m_i(t) - u_i(t) - f_i)$. Following Spiegel and Tookes (2013a), we let $\pi_i(t) + e^{\varsigma(t)} u_i(t) \equiv \hat{\pi}_i(t) = (\text{Revenue} - \text{Cost of Goods Sold})$. To keep the empirical problem manageable with the data at hand, the focus here is on the transition from α_i^* to α_i (for simplicity, set $f_i^* \equiv f_i$ and $s_i^* \equiv s_i$). Under the assumption that, for every firm i , $\alpha_i = k\alpha_i^*$, the profitability equation becomes

$$\pi_i(t) - e^{\varsigma(t)} u_i(t) = e^{\varsigma(t)} \left(((k - (k - 1)e^{-\psi t})\alpha_i^*)m_i(t) - f_i^* \right). \quad (14)$$

Using quarterly data on revenue and costs of goods sold, we use nonlinear least squares to estimate Equation (14) by IPO event. This gives estimates for $\psi, k, g, \alpha_i^*, \alpha_i$ and f_i^* . Recall that ψ is an industry level parameter that defines the transition rate from the pre-IPO α_i^* to the new α_i . The economic forces driving the transition and the associated "transition rate" should be broadly construed. It may be due to the IPO itself as in *H1*. Alternatively, the change may be due to external forces, per *H2*, *H3*, and *H4*.

3.2 Consumer responsiveness and competitive strength

Estimates for consumer responsiveness and competitive strength (ϕ and $\alpha_i s_i z$, respectively) are also required to estimate model-implied values. Recall that, because profitability evolves at the common rate ψ and given the common factor k scaling α_i^* relative to α_i , the transition from α_i^* to α_i does not affect $\alpha_i s_i z$. Letting $\bar{m}_i$ equal firm i 's steady-state market share, one can use the equation $dm = \phi(\bar{m}_i - m_i(t))dt$, which has a solution for $m_i(t)$ of

$$m_i(t) = \bar{m}_i + (m_i(0) - \bar{m}_i)e^{-t\phi}. \quad (15)$$

We use quarterly market share data to estimate Equation (15) for each IPO event. We use nonlinear least squares to obtain estimates for each firm's steady-state market share, $\bar{m}_i$, as well as for the industry parameter ϕ . The empirical identification comes from market share evolution across rival firms (to identify $\bar{m}_i$) and over time (to identify ϕ). Because the market shares of the n industry competitors must add to 1, there are $n-1$ independent observations in each quarter. To ensure that quarterly market shares do not add to one, we remove the smallest publicly traded rival in each period. We also exclude the IPO firm (this helps with the causality tests presented later). Estimation is done using the iterative Gauss-Newton method, with starting values specified for all unknown parameters. Note that we also obtain estimates for firm-specific competitive strength $\alpha_i s_i z$ via Equation (15), because, in steady state, $\bar{m}_i = 1 - \frac{n-1}{\alpha_i s_i z}$.

Equation (15) is estimated with data from both before and after the IPO event. We allow ϕ to vary pre- and post-IPO (ϕ_0 and ϕ , respectively) to account

for possible consumer behavior changes near the IPO event. Following Spiegel and Tookes (2013a), the only restriction that we impose is that ϕ_0 and ϕ are nonnegative and less than 25. In our quarterly estimation, this would correspond to a firm losing half its customers in just 2.5 days if it ceased all spending to maintain and expand its market share.

3.3 Data and summary statistics

The initial sample of IPO events is from Securities Data Corporation New Issues Database and includes the IPOs of U.S. publicly listed stocks from 1983 through 2012.¹⁵ There are 409 industries with IPO firms in CRSP/Compustat and for which the IPO firm is not the only firm in the Compustat 4-digit Standard Industrial Classification (SIC) code. Because we are interested in firm and industry heterogeneity, we estimate the model separately for each IPO event. The quarterly data for estimation include all rival firms that share the IPO firms' 4-digit SIC codes. Depending on the test, it is necessary to drop some IPO events. Table 1 displays the size of the database and the impact that various filters, which we describe in more detail below, have on the sample.

Starting with the initial sample of 409 IPOs, we remove utilities and financial firms (SIC codes 6000–6999 and 4900–4999) because utility regulation restricts interfirm competition, and the model is better suited for nonfinancial operating entities. Because the model assumes oligopolistic competition, we only include industries with fifty or fewer competitors.¹⁶ Restricting the sample in this way will help us capture the competitive effects of IPOs, if they are in the data. Doing so reduces the sample to 368 industries. All publicly traded rival firms for which we have market share data at the beginning of the estimation period are included. We do not include the IPO firm in the estimation. This is important because we will test whether parameter estimates based on rival firm performance are capable of describing the post-IPO performance of the IPO firm.

We begin the estimation 3 years prior to the IPO quarter. We estimate the model over horizons that include data through 3 and 5 years post-IPO, using a total of 24 and 32 quarters of data for each firm, respectively. Figure 1 shows a time line, using the 3-year horizon as an example.¹⁷

No model can hope to fit every industry, and this one is no exception. This paper's model is of a heterogeneous product oligopoly. One can think of many industries for which this is likely to be a poor approximation. Not surprisingly, the parameter estimates either fail to converge or converge to a corner solution

¹⁵ The IPO issue sample ends in 2012 to ensure a sufficient number of post-IPO quarters for estimating the model. The financial data from CRSP and Compustat are through December 31, 2014.

¹⁶ We chose fifty firms in an attempt to identify oligopolies, while still retaining as much data as possible. We are aware that this restriction is unlikely to be sufficient for identifying oligopolies. Our subsequent checks of parameter estimates for reasonableness (i.e., removing outliers) help us get closer to identifying industries for which the oligopoly model is and is not appropriate.

¹⁷ Data from quarters –12 through +11 (–12 to +19) relative to the IPO quarter 0 are used for the 3-year (5-year) in-sample estimation horizon.

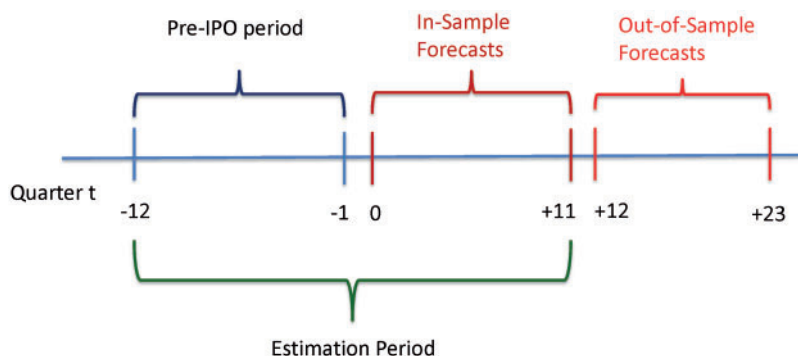


Figure 1
Estimation and event windows (3-year horizon example)

Figure 1 shows the window over which the model is estimated and the time horizons over which both in- and out-of-sample forecasts are constructed. The IPO occurs during quarter $t = 0$.

for some industries.¹⁸ Nevertheless, as summarized in Table 1, the success rate is reasonable. For the 3-year sample, the model converges for 53% of the industries with IPOs and 16% of all IPO events (217 industries and 741 IPO events, respectively); and for the 5-year sample, the model converges for 56% of industries and 19% of IPO events (227 industries and 855 IPO events, respectively). Even after applying the initial industry filters and requiring model convergence, we are able to analyze a much larger IPO sample than in HRR. They begin with an initial sample of 4,188 IPO events and use data on only 134 IPOs (3% of the initial sample) for their analysis.

Table 2, panel A, shows parameter estimates. The median estimated ψ ranges from 0.304 to 0.387. These imply that, following the median IPO event, between 26% and 32% of the transition from the old to the new profitability regime occurs in the first quarter. At the end of 4 quarters, the transition is between 70% and 78% complete. By the end of the second year between 91% and 95% of the transition has occurred. However, there is substantial cross-sectional variation in ψ .

Table 2 also provides estimates of the value of the profitability shock k , where $\alpha_i = k\alpha_i^*$. Across both of the estimation horizons, the median estimated value of the profitability shock k is less than 1. Consistent with prior work on the impact of IPOs on rivals, this implies that IPO events are typically followed by a reduction in industry-wide profitability per unit of market share. However, as in the case of ψ , there is substantial cross-sectional variation. The interquartile range for the estimated value of k using data over the 5 years following the IPO is 0.54 to 1.40. This implies that rival firms experience between a 46% drop

¹⁸ As mentioned in the Introduction, we do not claim that the model is suitable for all industries. For example, the model is not intended for industries for which $r < g$.

Table 1
Data filtering rules and total observations

Data requirement	In-sample, 3-year horizon				Out-of-sample, 3-year horizon			
	# 4-digit SIC codes	# IPO events	# rival firms	Firm-quarter observations	# 4-digit SIC codes	# IPO events	# rival firms	Firm-quarter observations
Initial sample of IPO events in SDC/Compustat	402	4,519	119,933	1,217,561	400	4,243	85,395	871,627
Remove financials and utilities	362	3,999	109,282	1,107,391	362	3,758	77,522	789,331
Require > 50 rivals	361	3,065	38,549	401,481	361	2,896	29,103	303,416
Model convergence	217	741	12,282	128,859	216	730	9,240	96,237
Trim at 1st and 99th percentile parameter estimates	211	692	7,820	93,018	208	670	6,464	73,433
Remove unreasonable profitability change estimates	211	692	7,820	91,351	208	670	6,460	65,160
Remove unreasonable value change estimates	211	687	5,535	62,497	206	664	5,204	46,148
In-sample, 5-year horizon					Out-of-sample, 5-year horizon			
Initial sample of IPO events in SDC/Compustat	402	4,519	120,619	1,829,046	387	4,102	69,252	1,124,478
Remove financials and utilities	362	3,999	109,918	1,661,385	349	3,637	62,964	1,016,012
Require > 50 rivals	361	3,065	37,737	612,853	348	3,106	24,497	408,809
Model convergence	227	855	14,232	227,583	225	828	9,102	151,635
Trim at 1st and 99th percentile parameter estimates	215	801	9,053	169,409	209	761	6,411	117,330
Remove unreasonable profitability change estimates	215	801	9,048	165,008	209	761	6,407	107,725
Remove unreasonable value change estimates	215	796	6,873	118,499	209	757	5,620	83,916

and a 40% increase in profit per unit market share (α_i) following an IPO.¹⁹ There are some IPO events for which estimated parameters are not reasonable (e.g., the maximum estimate for k for the 3-year estimation horizon is 305,260); however, most are quite plausible.

While industry profitability tends to decrease following IPOs, the industries are still growing. The median estimated quarterly real industry growth rate is near 0.6% per quarter under all specifications. This quarterly growth parameter varies within a reasonable range (e.g., based on the estimates using data for the 5-year window, we obtain estimates with an interquartile range of -1.0% to 2.1%).

The estimates for consumer responsiveness in the years surrounding IPO events are particularly informative. We find a striking decline in consumer loyalty at the time of the IPO. The median consumer responsiveness parameter pre-IPO (still using the 5-year estimation horizon as an example) of 0.040 implies that, for the median industry, it would take a competitor about 17 quarters to lose half of its customers if it completely stopped spending to attract them. For expositional purposes, call this the market share half-life.²⁰ The pre-IPO half-life of 17 quarters is more than 50% slower than in the broad industry sample in Spiegel and Tookes (2013a). Post-IPO, the values in the two studies look much more similar. Table 2 shows a median post-IPO ϕ of 0.129, which implies a market share half-life following the IPO of only 5.3 quarters. Economically, this seems to imply that industries with impending IPOs contain firms producing products that consumers view as relatively unique, at least when compared with the typical industry in the overall economy. Post-IPO, the industry transitions to a state in which consumers become less loyal to a particular firm's product and about as willing to substitute as elsewhere in the economy. These insights would be impossible to glean in the absence of the dynamic IPO model.

To sharpen the economic interpretation of the changes in profits per unit of market share (k) and consumer responsiveness (ϕ_0 versus ϕ), we can use the full set of estimated model parameters to obtain the projected long-run model-implied change in market value for the industry post-IPO. Table 2 shows the distributions of the parameters k , ϕ , ψ , and g . The final input to the firm value function is δ , the cost of capital minus the growth rate. We follow Spiegel and Tookes (2013a) and define δ as the long-run (1926 through period t) historical market risk premium plus the risk-free rate minus the long-run GDP growth rate. This value is identical for all firms for a given time period.²¹ Table 2, panel B, then uses these values to calculate the long-run change in rival firm value

¹⁹ Later in the paper, we use this cross-sectional variation to help us understand the relative importance of various explanations for the decline in rivals' post-IPO performance.

²⁰ The market share half-life is defined as $\log(2)/\phi$.

²¹ We make this simplification to reduce the noise in the industry-level cost of capital and growth estimates. We also do not include the $.5\sigma^2$ term, as doing so should not affect the cross-sectional tests.

Table 2
Summary statistics

Variable	Estimated parameter values										SD	N
	Mean	5th pctl	10th pctl	25th pctl	Med.	75th pctl	90th pctl	95th pctl	Min	Max		
	3-year estimation horizon											
ψ	0.674	0.065	0.101	0.187	0.387	0.882	1.561	2.169	0.000	9.021	0.806	741
k	1.115	0.223	0.361	0.646	0.893	1.359	1.977	3.007	0.000	305.260	14.754	741
g	0.006	-0.042	-0.028	-0.010	0.006	0.022	0.041	0.053	-0.152	0.243	0.033	741
ϕ_0	0.132	0.005	0.012	0.031	0.063	0.121	0.264	0.417	0.000	3.137	0.270	741
ϕ	0.451	0.017	0.038	0.116	0.256	0.531	1.050	1.710	0.000	4.775	0.594	741
5-year estimation horizon												
ψ	0.591	0.049	0.075	0.142	0.304	0.745	1.419	1.932	0.000	11.208	0.866	855
k	264	0.137	0.274	0.543	0.818	1.401	2.103	3.304	0.000	129.194	4.770	855
g	0.008	-0.035	-0.024	-0.010	0.006	0.021	0.039	0.053	-0.118	0.348	0.034	855
ϕ_0	0.102	0.002	0.006	0.017	0.040	0.085	0.181	0.317	0.000	3.797	0.285	855
ϕ	0.263	0.014	0.025	0.061	0.129	0.273	0.592	1.044	0.000	4.327	0.410	855

Notes: The model is estimated separately for each IPO event. Estimates for ψ , k , and g are from the profitability equation, given by Equation (4). Consumer responsiveness ϕ is estimated via equation (19) from Spiegel and Tookes (2013a). Estimates for pre- and post-IPO are labeled ϕ_0 and ϕ , respectively. IPO firm rivals are defined as those firms in the same Compustat 4-digit SIC code as the IPO firm. We use quarterly Compustat revenue and costs of goods sold data from the IPO quarter $t - 12$ through the beginning of quarters 12 and 20 following the IPO (3- and 5-year post-IPO estimation horizons, respectively). IPO events are taken from SDC from January 1, 1983 through December 31, 2012, in industries with fewer than fifty publicly traded competitors.

via a three-step process. First, plug the estimated parameters into Equations (8), (12), and (13), with t set to zero and infinity to obtain the model-implied pre- and post-IPO rival firm values, respectively. Second, calculate the ratio of the post- to pre-IPO industry market values. To reduce noise, firms that have an estimated value change in excess of a factor of 100 are dropped (post/pre of less than 0.01 or greater than 100). Third, for each IPO event, calculate rival firm median, value weighted mean, and equally weighted mean ratios.

As Equation (11) shows, firms will optimally respond to industry developments by changing their spending patterns. Thus, the equilibrium impact of $k < 1$ on firm value is smaller than it would be absent a strategic response. Based on the parameter estimates, post IPO the model forecasts the median firm in the median industry to lose just under 3% in value using 3- or 5-year horizon estimates with any of the filter rules. These magnitudes are smaller than, but broadly in line with HRR, who report average event period rival cumulative abnormal returns (CARs) during days -10 to +1 relative to the IPO date of negative 68 basis points and a CAR of negative 187 basis points during days -10 to +20.²² This may seem surprising because the estimated value of k is between 0.8 and 0.9 (i.e., a profitability decline of 1,000 to 2,000 basis points) for the median IPO in Table 2. Competition for market share reconciles the seeming contradiction. Before the IPO, suppose one unit of market share (a customer) is worth \$10. Firms will react by spending up to \$10 in customer acquisition activities to draw in additional units of market share. After the IPO, the same unit of market share sees its value reduced to \$8.50. Firms react by reducing their customer acquisition spending accordingly. The overall impact on profits is just the degree to which the value of k alters what amounts to a firm's local monopoly over the consumers that prefer its product.

That the empirical model fails to converge for a number of industries naturally raises the question of whether these industries are somehow "special." Table 3 compares the industry sample in which convergence did and did not take place. Overall, the differences tend to be economically small. For example, consider the IPO firm's size (likely one of the more important variables). In the no convergence sample, these firms had an average market share of 5.4%. By contrast, in the convergence sample their market share was 3.6%. While statistically different, the economic difference does not appear to be consequential to the general conclusions one might draw from the two samples. Other statistics, like post-IPO sales growth, are statistically indistinguishable.

3.4 Causes and consequences of IPOs

The hypothesis list presented in Section 2.1 makes it possible to classify an IPO's causes and consequences based on a combination of estimated

²² Although the HRR paper focuses on 134 large IPOs at the 2-digit industry level, their results are useful as a benchmark. These short-run results are broadly consistent with the median value changes that we find. HRR credit this change to the competitive advantages the IPO firm sees from going public. The next section uses the model's forecast errors to understand why rival firm values are changing at all.

Table 3
Industry comparison: Convergence versus no convergence

3-year estimation horizon			Difference	t-stat	p-value
Converged	No	Yes			
Profitability, $t = -1$	0.169	0.172	0.003	0.65	.52
Industry size (\$000), $t = -1$	\$4,591	\$3,880	-\$711	(1.56)	.12
Number of firms, $t = 0$	18,888	20,146	1,258**	2.20	.03
IPO firm market share $t = 0$	0.054	0.036	-0.018***	-3.87	.00
3-year post-IPO ch. profitability	-0.011	0.000	0.012***	2.89	.01
5-year post-IPO ch. profitability	-0.018	-0.013	0.005	1.15	.25
Sales growth (3 years post-IPO)	0.107	0.095	-0.013	-0.64	.52
Sales growth (5 years post-IPO)	0.069	0.076	0.007	0.94	.35
Number of industries	151	217			
5-year estimation horizon					
Converged	No	Yes	Difference	t-stat	p-value
Profitability, $t = 1$	0.169	0.181	0.012***	2.72	.01
Industry size (\$000), $t = 0$	\$4,591	\$4,023	\$568	(1.30)	.19
Number of firms, $t = 0$	18,888	20,214	1,326**	2.45	.01
IPO firm market share $t = 0$	0.054	0.037	-0.017***	-3.97	.00
3-year post-IPO ch. profitability	-0.011	-0.007	0.004	1.21	.23
5-year post-IPO ch. profitability	-0.018	-0.015	0.003	0.75	.45
Sales growth (3 years post-IPO)	0.107	0.098	-0.009	-0.50	.62
Sales growth (5 years post-IPO)	0.070	0.081	0.011	1.60	.11
Number of industries	141	227			

Notes: This table presents average characteristics of industries in which the model does and does not converge. *Profitability* is defined as the average (across all firms in the 4-digit SIC code) quarterly revenues minus cost of goods sold, divided by total firm assets in quarter $t - 1$ relative to the IPO. *Industry size* is defined as the sum of quarterly revenue across all firms in the 4-digit SIC code during the quarter prior to the IPO event. *Number of firms* is the number of Compustat firms in the industry during the quarter of the IPO. *IPO firm market share* is the IPO firm's quarterly sales divided by the sum of quarterly sales in the industry during the quarter of the IPO event. *3-year post-IPO ch. profitability* is the industry average 3-year change in profitability beginning in quarter $t - 1$ relative to the IPO event. *5-year post-IPO ch. profitability* is the average 5-year change in profitability beginning in quarter $t - 1$ relative to the IPO event. *Sales growth* is the average annual growth in industry sales, beginning at quarter $t - 1$ relative to the IPO event and measured over 3- and 5-year horizons. * $p < .1$; ** $p < .05$; *** $p < .01$.

model parameters and out-of-sample forecast errors for the firms in the affected industries. For example, under $H1$, the post-IPO decline in industry performance arises from the IPO firm's improved competitive advantage. To see when this applies, we first identify industries in which we observe either a significant decline in profitability per unit market share ($k < 1$) and/or an increase in consumer responsiveness ($\phi - \phi_0 > 0$). We then use the out-of-sample profitability and value change forecast errors to test for whether the IPO firm is outperforming its industry. By comparison, to examine whether the data are more consistent with the hypothesis that an IPO is indicative of deteriorating industry conditions ($H2$ and $H3$), we also identify industries with $k < 1$ and/or $\phi - \phi_0 > 0$. However, for the data to support these hypotheses, the forecast errors for the IPO firms and rivals must not be different from each other.

Before discussing the results of the forecast error tests, it is useful to discuss their efficiency. We use parameter estimates from a first-stage model to generate a series of predicted values that we, in turn, use to calculate forecast errors. We choose to estimate the first-stage model separately for each event rather than conducting all of our analysis in one step using GMM, for example. GMM is more efficient in theory but can be less feasible in our setting, given the industry-, event-, and firm-level heterogeneity that we aim to capture in the parameter estimation. Still, it is important to recognize that the forecast errors that we analyze contain noise from the first-stage estimation. Given the nonlinear structure of the first-stage model, this noise may have asymmetric effects on the distribution of predicted values. Later in the paper, we address potential issue in two ways. First, we examine out-of-sample forecasts to understand the extent to which the model fits the data. Second, we examine the impact of potentially noisy parameter estimates by simulating a set of new parameters using the distributions from the point estimates that we obtain in the first-stage estimation. We then run out-of-sample tests using the simulated values.²³ Third, we conduct Harvey and Collier (1977) model misspecification tests. All three of these checks of the data show that model-implied values indeed have predictive power and can be used to understand the changes occurring near IPO events. Section 3.8 describes these tests in detail.

Table 4 summarizes the results of the forecast error tests. We find strongest support for the hypothesis that IPOs occur when industry-wide fundamentals

²³ We thank an anonymous referee for encouraging this line of analysis.

²³ We supplement the analysis of the competitive effects of IPOs by examining the estimated coefficient from a regression of estimated changes in profitability ($\log(k)$) on the log of the number of IPOs in the industry during quarter t . For the $k < 1$ subsample, it is statistically insignificant (-0.093, with a t -value of -0.78 using the 3-year horizon and -0.036, with a t -value of -0.27 using the 5-year horizon). For the $k > 1$ subsample, the coefficient is also insignificant (-0.021, with a t -value of -0.36 using the 3-year horizon and -0.33, with a t -value of -0.43 using the 5-year horizon). Thus, the evidence does not support the hypothesis that more IPOs cause larger declines in profitability.

²³ More than half of IPO events (53.7%, or 398 of 741) are consistent with $H2$ or $H3$ using a 3-year horizon. More than half of IPO events (52.9%, or 452 of 855) are also consistent with $H2$ or $H3$ using a 5-year horizon.

Table 4
Categorizing an IPO's causes and consequences

Hypothesis	Top findings: 3-year horizon Bottom findings: 5-year horizon	Median IPO mkt. share in %
<i>H0 (Null): Firm-specific reasons for going public and/or outside of the model, without competitive implications</i>	32.9% (244/741) of the events are consistent with <i>H0</i> 35.2% (301/855) of the events are consistent with <i>H0</i>	1.02 1.36
<i>H1: Competitive effects of IPOs</i> ²⁶	6.2% (46/741) of the events are consistent with <i>H1</i> 7.8% (67/855) of the events are consistent with <i>H1</i>	1.74 1.43
<i>H2: Industry-wide shift in consumer loyalty for products</i>	48.2% (357/741) of the events are consistent with <i>H2</i> 47.1% (403/855) of the events are consistent with <i>H2</i>	1.39 1.74
<i>H3: Industry-wide shift in profitability</i>	37.9% (281/741) of the events are consistent with <i>H3</i> 40.6% (347/855) of the events are consistent with <i>H3</i> ²⁷	1.90 1.62
<i>H4: IPOs are associated with improving industry conditions</i>	7.0% (52/741) of the events are consistent with <i>H4</i> ; 4.1% (35/855) of the sample of events are consistent with <i>H4</i>	1.36 0.95

Notes: This table lists several mechanisms capable of explaining the main findings in the paper, along with the frequency with which we find empirical support. To test each hypothesis, we define statistical significance at the 10% level. Note that some hypotheses are not mutually exclusive. Therefore, the percentages reported in the findings column do not necessarily add to 100%. *k* is the post-IPO change in industry profitability per unit market share, where *k*= 1 indicates no change. "Median IPO mkt. share in %" is the average (median) IPO firm's market share at the time the firm goes public for industries with parameter estimates fitting a particular hypothesis.

are in decline (i.e., the “canary hypotheses,” $H2$ and $H3$). Using the estimates from the 5-year estimation horizon as an example, 52.9% of the observations are consistent with deteriorating industry conditions and no overperformance (i.e., positive forecast errors) of IPO firms or their industry rivals. While far less common, there is also some evidence of competitive effects of IPOs. We find evidence consistent with the idea that rival firms suffer due to a stronger IPO firm in 7.8% of all events. However, not all events are followed by clear declines in industry fundamentals. Contrary to what $H1$ to $H3$ predict, in 4.1% of IPO events, we find evidence of improving, rather than deteriorating, industry conditions. In another 35.2% of the events, we find evidence consistent with the null hypothesis $H0$, in which changes are occurring for reasons outside of the model. Most of these are cases in which industry conditions are changing in ambiguous ways (e.g., k significantly greater than 1, but $\phi - \phi_0$ significantly less than zero).²⁴

The last column in Table 4, which shows the median IPO firm market share for all events that are consistent with a given hypothesis, reveals a couple of interesting patterns that may offer additional insights into the question of what triggers the IPO in the first place. HRR restricted their sample to the largest firm in a 2-digit SIC code to go public in a 12-year window (6 prior and 6 after). As a result, the IPO firms in their sample are much larger than most of their competitors. HRR’s table II reports that the average IPO firm has a market capitalization that is 70% larger than the industry average. However, as the IPO market share column in our Table 4 indicates, IPO firm market shares are not particularly large in industries where the parameter estimates support the hypothesis advanced in HRR that rivals suffer from the increased strength of new public firms ($H1$). In fact, the median market shares are similar to the market shares of IPO firms in cases where we observe industry-wide shifts ($H2$ and $H3$). If a firm has remained private until it is particularly large, the question arises as to what actually sparks the decision to go public. The estimates in Table 4 indicate that, for such firms, it is typically a change in the overall industry that pushes the decision, rather than the expectation that, by becoming public, the firm’s competitive position will become substantially stronger.²⁵

²⁴ We do not have enough information to run forecast error tests for many events, which are required for consistency with some of the hypotheses listed in Section 2.1 (e.g., $H2$ requires positive forecast errors for the IPO firm). The missing data are due to filtering implausible estimates, as described in Table 1 and in Appendix Section A.4. To ensure that the percentage frequencies in Table 4 are not due to forecast error data availability, we repeat the analysis, but, for an event to be included in the sample, we require data on forecast error correlations between the IPO firms and their rivals. Results are in the Online Appendix Table A.3 and are qualitatively similar to those in Table 4.

²⁵ In interpreting the results in Table 4, one might be concerned that the power of the forecast error test is insufficient to detect a true difference in IPO firm performance. To examine this potential issue, we performed tests using both in- and out-of-sample data for all firms in the industry, in which we estimated panel regressions of actual firm performance on model-implied performance, plus an IPO firm dummy. The IPO firm dummy is never significantly positive (i.e., no overperformance). We then repeated the analyses but replaced actual IPO firm

Table 5
Causes and consequences of IPOs: Bayesian updating exercise

Hypothesis	3-year horizon				5-year horizon			
	Mean	SD	95% HPD interval		Mean	SD	95% HPD interval	
<i>H0</i> : Outside of the model	0.3289	0.0177	0.2945	0.3630	0.3513	0.0156	0.3203	0.3817
<i>H1</i> : Stronger IPO firm	0.0625	0.0088	0.0448	0.0794	0.0788	0.0090	0.0611	0.0954
<i>H2/H3</i> : Deteriorating industry conditions	0.5379	0.0194	0.4996	0.5730	0.5285	0.0166	0.4960	0.5630
<i>H4</i> : Improving industry conditions	0.0707	0.0093	0.0539	0.0903	0.0415	0.0066	0.0292	0.0543

Notes: This table presents results from a Bayesian analysis of the data in Table 4. Under *H0*, the industry changes that we observe following the IPO occur for reasons outside of the model. Under *H1*, the changes occur because the IPO firm is a stronger competitor. Under *H2* and *H3*, which are considered together, the changes occur because of deteriorating fundamentals at the industry level (i.e., declining profitability, increasing consumer responsiveness or both). Under *H4*, industry fundamentals are improving. We set equal priors for the four (mutually exclusive) outcomes. We then use data from Table 4 to calculate posterior distributions. The table shows the posteriors after 1,000 simulations. The HPD is the highest posterior density interval.

The findings in Table 4 show that more events fit the “industry trend” explanation than the other alternatives. We can use Bayesian analysis to provide more formal interpretation of the data. In particular, we can ask how a Bayesian would update her prior, given the data in Table 4. We set equal priors for four mutually exclusive explanations for the industry changes that we observe following IPOs, as discussed in Section 2.1. This analysis combines *H2* and *H3*, both of which imply that changes occur because of deteriorating fundamentals at the industry level although from potentially different sources (i.e., declining profitability, increasing consumer responsiveness, or both). We then use data from Table 4 to calculate posterior distributions. Table 5 shows the posteriors after 1,000 simulations.²⁶ The 95% highest posterior density interval for the deteriorating industry conditions hypothesis ranges from 0.49 to 0.57. The posteriors for the other explanations are all considerably smaller.²⁷ The findings in Tables 4 and 5 provide strongest support for the “canary” hypothesis, but it is also important to emphasize a Bayesian would likely assign a weight in the range of .044 to .095 to the “competitive IPO” hypothesis.

performance values with ones that were 1%, 2%, and 3% greater than their true values (i.e., we generated false overperformance). We then estimated 12 sets of regressions: in-and-out of sample tests for the 3- and 5-year horizons using 1%, 2%, and 3% overperformance levels. For profitability, overperformance was detected in all 12 cases. For value change, we rejected the null in ten cases: all six of the out-of-sample tests and four of the six in-sample tests (i.e., in all but the 1% overperformance case). Thus, the evidence that IPO firms do not outperform rivals is not likely to be due to low statistical power in the tests.

²⁶ We use the SAS MCMC procedure to obtain posterior distributions. We thank Feng Chen of SAS for statistical support.

²⁷ Alternatively, we could take a frequentist approach and ask what the results in Table 4 would look like if we vary the significance thresholds used to define statistical significance. In Online Appendix Table A.2, we report results using 5% and 1% thresholds.

3.5 Panel regressions

Another way to clarify the overall interpretation of the estimated parameter sets is to ask how the changes in the industry relate to the timing of a firm's IPO. As noted earlier, this paper finds that consumer responsiveness ϕ tends to increase following going-public events. We hypothesize that reduced consumer loyalty can encourage firms to go public by reducing the value of maintaining competitive secrecy via private financing (*H2*). One potential implication of this is that an increase in the willingness of consumers to switch products should lead firms to go public earlier (i.e., when their market shares are smaller), when the anticipated value of ϕ is larger.²⁸

To test whether an increased willingness of consumers to switch products encourages smaller firms to go public, we run a panel regression with the IPO firm's market share at issuance as the dependent variable. The independent variable of interest is the change in consumer responsiveness, $\phi - \phi_0$. The pre-IPO consumer responsiveness variable ϕ_0 is estimated with data from the 12 calendar quarters prior to the IPO. As shown in Figure 1, the post-IPO value of ϕ is then estimated with data from the 12- and 20 quarters beginning at IPO quarter $t = 0$, depending on whether the tests are based on 3 or 5 year horizons, respectively. While we hypothesize that an increase in consumer willingness to switch products is associated with smaller firms going public, other variables may be at work to produce the observed pattern. The obvious alternatives are those used in other studies of rivals' post-IPO performance. In particular, we control for variables used by HRR and CL in the analysis of IPO firm market share.²⁹ To simplify the subsequent exposition, Table 6 defines all variables.

Table 7 presents results from regressing the $t = 0$ market share of the IPO firms on $\phi - \phi_0$, 4-digit SIC code fixed effects, industry sales growth over the previous year, industry size, and control variables from HRR and CL. Given the 4-digit SIC code fixed effects, the coefficient on $\phi - \phi_0$ captures, within industry, the impact of larger changes in consumer responsiveness on the size of the IPO firm at issue. Five of the six estimated coefficients on $\phi - \phi_0$ in Table 7 are negative and, when the 3-year estimation horizon is used, all are negative and both statistically and economically significant. For example, consider the estimated coefficient of -0.0099 from the first column of Table 7. In the median industry, the change in ϕ is -0.168 post-IPO. Multiplying these two values together implies that, following the i^{th} IPO, if the industry sees the median change in ϕ , then IPO firm $i + 1$ will go public with a market share that is about 17 basis points smaller than the share at which the i^{th} firm did. This is substantial considering the sample average at-issuance market share is 116 basis points. The

²⁸ We formalize this intuition in the Online Appendix, in which we endogenize the IPO decision. We find that product homogenization (higher ϕ) leads to lower profits and simultaneously reduces the optimal size at which a firm should go public.

²⁹ Following CL, we only include industries where the CSM estimates are negative. This makes the sample size smaller for the sample in which we include the CL variables.

Table 6
Variable definitions

Variable	Definitions and estimation procedures
Model parameters	
$\Delta value$	Defined as the log of the value in quarter t , divided by the value in period $t-1$. Model-implied values are given by the slow-transition model value functions in Equations (12) and (13)
$\Delta profit$	Profit changes are defined as profit in period t minus profit in period $t-1$, divided by the $t=0$ value of assets. Model-implied profitability is based on the profitability equation given in Equation (9) and estimates for ψ , k , g , α_i , and f_i derive from it
$lag_dependent_var$	One-quarter lag of $\Delta value$ or $\Delta profit$
ϕ	Consumer responsiveness is estimated from Spiegel and Tookes (2013a equation (19)). Estimates for pre- and post-IPO are labeled ϕ_0 and ϕ , respectively
δ	The discount rate net of growth variable δ is defined as the long-run (1926 through period t) historical market risk premium plus the risk-free rate minus the long-run GDP growth rate. Market-wide δt is identical for all firms
HRR variables	
log_at	Natural log of total assets
ind_mb	Median industry market-to-book ratio in the previous year
$underprice$	The annual level of underpricing in a given year t
age	Natural log of the number of years since the firm's first trading day in CRSP
ipo_dummy	Dummy equal to 1 if the quarter occurs in the IPO year or in years 1, 2, or 3 following the IPO
CL variables	
CSM	Estimated using 20 rolling quarters of historical data firm-by-firm. It is defined as $CSM_i = corr \left[\frac{\Delta \pi_i}{\Delta S_i}, \Delta S_{-i} \right]$, where ΔS_i equals the change in firm i 's sales and ΔS_{-i} equals the change in the sales of its rivals. Industry CSM is defined as the median of the firm-by-firm estimates
$d_uncertain$	Standard deviation of seasonally adjusted sales growth. Calculated using 20 rolling quarters of historical data. Following Chod and Lyandres (2011), we focus only on the subsample of industries with negative CSM
$s_uncertain$	Systematic component of demand uncertainty, defined as the ratio of the variance of the predicted values from a regression of seasonally adjusted industry sales growth on the seasonally adjusted sales growth of all Compustat firms

Notes: Model parameters are estimated using data from all public firms with the same 4-digit SIC code as an IPO firm. Revenue and cost of goods sold come from quarterly Compustat data from the IPO quarter t to the beginning of quarters $t+12$ and $t+20$ (3- and 5-year post-IPO estimation horizons, respectively).

weaker results for the 5-year horizon line up with the intuition that changes in consumer demand are easier for firms to forecast over shorter periods. Overall, the evidence in Table 7 sharpens the paper's general conclusion: IPO firms tend to go public in response to impending commoditization in the product market.

The findings presented so far do not rule out the possibility that the IPO firm chooses to go public at the expense of the entire industry in order to avoid larger losses than it would incur if it remained private. This is a variant of $H1$ that may be misclassified as falling under $H2$ or $H3$. We can use what we know about the strategic alternatives available to incumbents to help clarify

Table 7
IPO firm size at issuance and change in consumer responsiveness

	Estimate	t-value	Estimate	t-value	Estimate	t-value
Intercept	0.0000	0.00	0.0000	0.00	0.0000	0.00
$\phi - \phi_0$	-0.0099***	-3.48	-0.0101***	-3.55	-0.0134**	-2.47
<i>g</i>	0.0693	1.05	0.0592	0.85	-0.0045	-0.05
<i>ind_size</i>	-0.0126	-7.32***	-0.0125***	-7.12	-0.0162***	-3.88
<i>ind_mb</i>			0.0028	0.92	0.0006	0.09
<i>underprice</i>			-0.0139	-1.31	0.0167	0.68
<i>CSM</i>					0.0810	1.17
<i>d_uncertain</i>					-0.0241	-0.69
<i>s_uncertain</i>					0.0150	0.68
<i>N</i>	741		741		363	
Adj. <i>R</i> -squared	6.061%		6.271%		7.641%	

	Estimate	t-value	Estimate	t-value	Estimate	t-value
<i>Intercept</i>	0.0000	0.00	-0.0000	-0.00	0.0000	-0.00
$\phi - \phi_0$	-0.0016	-0.36	-0.0017	-0.36	0.0008	0.16
<i>g</i>	0.0575	1.49	0.0467	1.14	0.0575	1.24
<i>ind_size</i>	-0.0058***	-2.85	-0.0057**	-2.07	-0.0085***	-2.88
<i>ind_mb</i>			0.0037*	1.78	0.0063***	3.07
<i>underprice</i>			-0.0145	-0.89	0.0218	1.03
<i>CSM</i>					0.0000	0.00
<i>d_uncertain</i>					-0.0090	-0.31
<i>s_uncertain</i>					0.0318	1.53
<i>N</i>	855		855		425	
Adj. <i>R</i> -squared	1.011%		1.386%		2.639%	

Notes: This table shows results from regressing the IPO firm’s market share on the change in consumer responsiveness ($\phi - \phi_0$) and industry fixed effects, industry growth, and control variables based on HRR and CL. See Table 6 for variable definitions. The variable labeled $\phi - \phi_0$ is defined as the difference between ϕ estimated using data from the 12 and 20 quarters following the IPO event (3- and 5-year horizons, respectively) and ϕ_0 , estimated using data from the 12 calendar quarters preceding the IPO. All *t*-statistics are based on standard errors clustered by year. Four-digit SIC code fixed effects are estimated, but not reported. * $p < .1$; ** $p < .05$; *** $p < .01$.

the interpretation. Suppose that, absent the IPO, the whole industry would not see its value drop. But if the firm contemplating the IPO remains private, its value will decline by *X*%. The IPO, however, will cause it and its industry all to decline by *Y*%. Assuming $Y < X$, then our tests would indicate the IPO firm does not outperform the industry as a whole, yet it would have an incentive to trigger the industry’s decline in order to maximize its own value. In theory, this could be the case but it raises the question of why larger rivals do not simply purchase more of their private competitors before they go public.

Table 8 shows that IPO firms are very small relative to their largest rivals. While there is a robust market for acquiring private firms, Table 8 suggests that if the IPO firm is the cause of an industry’s subsequent decline, we should see much more of this type of activity. The median value of the largest firm across industries in our sample is more than 20 times the size of the IPO firm. Restricting attention to just those industries where post-IPO profits decline (those where purchasing the target would prevent the value losses, under the hypothesis that IPOs cause the industry to change,) results in similar ratios. HRR report in their Figure 1 an average long-run post-IPO value decline per industry of just under 2%. A 20-to-1 size difference means that the largest company in an

Table 8
Market value ratio per industry: Largest firm to IPO firm

Group (3-year horizon)	10th pctl	25th pctl	Median	75th pctl	90th pctl
All industries	2.69	6.60	22.29	71.29	242.14
Profitability declines post-IPO	3.25	7.55	24.79	86.51	298.14
Group (5-year horizon)	10th pctl	25th pctl	Median	75th pctl	90th pctl
All industries	2.69	7.02	21.84	69.29	224.45
Profitability declines post-IPO	3.26	8.15	23.69	78.45	258.75

Notes: This table shows the distribution of the ratio of the enterprise value of the largest firm in an industry to that of the IPO firm. All IPO industries are included in the tabulation. The row labeled “Profitability Declines Post-IPO” refers to those industries in which aggregate profitability declines after the IPO is conducted.

industry can purchase the still-private firm, lose almost 40% on the transaction, and remain better off. At the 75th percentile of this size ratio, the largest firm is 70 times bigger than the IPO firm. In this case, the larger company can purchase and shut down the nascent IPO firm and retain value relative to letting the IPO trigger a decline in industry value. Now, consider again the argument that IPOs trigger declines in industry value and are conducted because the IPO firm would otherwise be even worse off if it remained private. If true, then one now has to ask why many more of the observed IPOs in the data were not short-circuited by vastly larger rivals through acquisitions of firms prior to their going public.

3.6 Model verification and comparison with prior models

Given our reliance on model estimation errors to infer causality, it is useful to ask how well the model fits the post-IPO performance data. Does it produce results superior to those of models that employ simple linear structures to explain rivals’ post-IPO performance? The HRR, CH, and CL models all analyze post-IPO data of rival firms using regressions unrestricted by a model. The next sections reveal that the forecasts of this paper’s structural model are superior in a number of dimensions to the independent variables that other papers use.

3.7 Out-of-sample tests

Dynamic structural models have the potential to advance beyond the limits of static models by, in part, offering a way to predict future events. To see how the model in this paper performs in real time, we conduct a set of out-of-sample tests. In them, a period $t + 1$ forecast is made solely on the basis of data available as of period t . These tests also allow one to check for model misspecification. A misspecified model may produce in-sample parameter estimates, which may even be “statistically significant.” However, out-of-sample, it will perform poorly if model itself does not represent the true process under consideration. In what follows, the tests compare how well the model estimated here does against both an autoregressive model and forecasts based on variables that others have used in the literature linking IPOs to an industry’s subsequent performance. As will be seen, the dynamic model in this paper does quite well at explaining the out-of-sample evolution of the data.

A true forecast uses only data available to investors at time t to make a projection about some value as of time $t+1$ or further into the future. In a dynamic model, this is a straightforward exercise. Table 9 begins with the model-implied value and profitability changes based on rolling out-of-sample parameters and market shares estimated with data through period $t-1$. The period-ahead market share forecasts are generated from regressions of market share changes on: model-implied changes in market shares (based on Spiegel and Tookes 2013a, equation (19)); 1-quarter lagged changes in market share; and 1-quarter lagged market share levels (using data through $t-1$). The out-of-sample tests begin at 12 and 20 quarters following the IPO (3- and 5-year horizons, respectively). From Table 9, the model-implied forecasts are clearly significant in all of the univariate regressions, particularly at the 3-year estimation horizon. At the 5-year horizon (which examines data from 5 years to 10 years post-IPO), the coefficients are still significant, but only at the 10% level. This might be expected; as other developments are likely to affect the industry by year 10 following an IPO.

In Table 9, we also ask how well the dynamic model performs compared to the other variables in the literature. Relative to a dynamic model, constructing a set of forecasts for a static model is challenging. A static model by design combines the current period's parameter and explanatory variables to produce an expected value of the current period's dependent variable. To work around this problem, we take the following approach. For every period t , we use data through period $t-1$ to estimate regressions of profitability and value (levels) on the variables from HRR. We then interpret the period $t-1$ residual as the predicted change in value from period $t-1$ to period t . The basic idea is that if the HRR variables imply that actual profitability in period $t-1$ is higher than predicted, we should expect profitability to decrease in the next period (i.e., the HRR residuals are expected to have a negative sign in period-ahead predictive regressions). We construct similar predictions based on the measures proposed by CL. The right-hand-side columns of Table 9 present the results. In the case of value changes (panel A), the model-implied changes are the only statistically significant predictor. In the case of profitability, the model-implied changes as well as the changes implied by the HRR and CL variables are statistically significant, with the expected signs. These out-of-sample tests support our use of the forecast errors to shed light on the forces driving rivals' performance post-IPO.

A second potential concern with the two-step nonlinear estimation approach that we use is that the noise in the first-stage estimation can affect inference. To examine this possibility, we use the point estimates and the variance-covariance matrix of the estimates obtained in Table 2 to simulate new parameters that we can use to predict performance. We conduct 100 simulations and then run the same out-of-sample tests shown in Table 9 using the simulated data. Table 10 shows the distribution of the coefficients on model-implied changes parameter estimates (as well as the HRR and CL variables). There are three important

Table 9
Does the model have predictive power? Out-of-sample tests

Est. horizon	3-year		5-year		3-year		5-year		3-year		5-year	
	Est.	t-val.	Est.	t-val.	Est.	t-val.	Est.	t-val.	Est.	t-val.	Est.	t-val.
A. Dependent variable = Δ value												
Intercept	0.0048	0.53	0.008	0.92	0.0000	0.00	0.0000	0.00	0.000	0.00	0.0000	0.00
model_Δ	0.0258***	3.61	0.014*	1.67	0.0160*	1.83	0.0319***	4.05	0.026***	3.80	0.0409***	4.39
HRR					-0.0117	-0.95	-0.0117	-0.31	-0.014	-0.43	-0.0174	-0.47
CL									-0.025	-0.95	-0.0137	-0.83
N	46,148		83,916		45,933		80,298		24,470		44,119	
Adj. R-squared	0.096%		0.030%		0.150%		0.103%		0.260%		0.204%	
B. Dependent variable = Δ profit												
Intercept	0.0051**	2.34	0.0062***	2.61	0.0057**	1.97	0.0036**	1.60	0.0000	0.00	0.0000	0.00
model_Δ	0.0879***	4.83	0.0656***	3.90	0.0844***	4.77	0.1029***	5.41	0.0777***	3.48	0.0973***	3.93
HRR					-0.060***	-3.05	-0.0560***	-2.40	-0.0094	-0.34	-0.0353	-1.24
CL									-0.0435*	-1.90	-0.0203*	-0.83
N	65,160		107,725		64,147		102,273		34,861		56,669	
Adj. R-squared	0.597%		0.292%		0.957%		0.808%		0.822%		0.674%	

Results are from period-ahead predictive regressions of quarterly changes in rival firm value (Δ value) and profitability (Δ profit) on model-implied changes ($model_{\Delta}$), as well as the predicted changes based on the explanatory variables from Hsu, Reed, and Rocholl (2010) and Chod and Lyandres (2011). The predictive regressions are estimated using out-of-sample data from quarters 20 to 40 (5-year horizon) and from quarters 12 to 24 (3-year horizon). The period-ahead market shares used in the estimation are forecast from regressions of market share changes on model-implied changes in market shares, 1-quarter-lagged changes in market share, and 1-quarter-lagged market share levels. HRR and CL represent residuals from regressions using the HRR and CL variables as defined in Table 6. We use data through period $t-1$ to estimate regressions of profitability and value (levels) on the variables from HRR and CL. The period $t-1$ residual is the predicted change in value or profitability from period $t-1$ to period t . All regressions are pooled, with standard errors clustered by IPO event and calendar quarter end date. IPO event fixed effects are included in the HRR and CL regressions. For consistency with Chod and Lyandres (2011), only industries in which CSM is negative are included in the regressions in which the CL variables are included. * $p < .1$; ** $p < .05$; *** $p < .01$.

observations. First, for the profit change regressions, the estimated coefficients on the model-implied values are very similar to those in Table 9. Second, the estimated coefficients on the model-implied values in the value-change regressions are much smaller than those in Table 9. This is not surprising, given that the value change estimation is highly nonlinear. Third, and finally, the model *still* has significant predictive power, even after the noise in first-stage estimation is considered.³⁰

In addition to examining the significance of various regression coefficients, another way to compare the candidate variables is to perform a model selection analysis. Table 11 uses the Schwarz-Bayesian information criterion to do this. In these tests, the model forecast is always first. No other single variable performs as well. In fact, the test suggests excluding all the other regressors in one or more specifications. This is true of the lagged dependent variable as well, and it highlights the fact that there is little support for the alternative hypothesis that the value and profitability changes that we observe are predictable simply because they follow first-order autoregressive processes.

Harvey and Collier (1977) tests provide another way to see whether the model's estimates are driven by misspecification errors. We conduct these tests on the entire Compustat database, using data from 1995 to 2015. As in the IPO tests, utilities and financial firms are removed, as are industries with fewer than 3 and more than 50 competitors. We test for model misspecification across firms in two ways. The first tests follow the original Harvey and Collier (1977) paper, in which the authors perform a t -test on recursive residuals. We estimate the model using data to time t . We use these parameter estimates to create and record forecast errors for time $t+1$. We then repeat this procedure, using data up to time $t+1$ to create forecast errors for time $t+2$, and so on, through to the end of the period covered by the data. If there is no misspecification, then setting the size of the test to, say, 5% should result in a rejection of the model for about 5% of all firms. Generally, we find that the rejection rate is only about 2% larger than the test's size.³¹ This is true both overall and across firm sizes, with the exception of the very largest companies. For firms with market shares over 10% in the early period, the rejection rate can be up to 8% higher than the test size using a 10% p -value. The second set of tests accounts for the fact that industry structure is likely to change over very long horizons and uses rolling estimates of either 3 or 5 years (similar to the tests conducted with the IPO data). In this case, the rejection rates tend to be somewhat lower.

³⁰ Note that the sample size in Table 9 is larger than that in Table 10. This is because all of the Table 9 analysis is conducted conditional on having usable parameters for Table 9. Once we run simulations, some of the simulated data are discarded according to the same filters that are applied to the original data.

³¹ Online Appendix Table A.4 provides a detailed description of the Harvey-Collier tests and results.

Table 10
Does the model have predictive power? Out-of-sample tests using simulated data
A. 3-year horizon

Value changes		Simulated distributions									
		Parameter	Mean	SD	Max	90th pctl	75th pctl	Median	25th pctl	10th pctl	Min
1	1	<i>Intercept</i>	0.0073	0.0007	0.0092	0.0082	0.0078	0.0073	0.0068	0.0064	0.0056
1	1	<i>model_Δ</i>	0.0066	0.0027	0.0152	0.0103	0.0086	0.0064	0.0045	0.0031	0.0001
2	2	<i>Intercept</i>	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2	2	<i>model_Δ</i>	0.0079	0.0037	0.0195	0.0129	0.0105	0.0076	0.0059	0.0032	0.0008
2	2	<i>HRR</i>	(0.0150)	0.0074	0.0007	(0.0058)	(0.0098)	(0.0144)	(0.0211)	(0.0249)	(0.0356)
3	3	<i>Intercept</i>	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
3	3	<i>model_Δ</i>	0.0082	0.0047	0.0226	0.0142	0.0106	0.0082	0.0056	0.0016	0.0017
3	3	<i>HRR</i>	(0.0245)	0.0113	(0.0009)	(0.0098)	(0.0173)	(0.0236)	(0.0323)	(0.0387)	(0.0530)
3	3	<i>CL</i>	0.0123	0.0130	0.0389	0.0282	0.0215	0.0135	0.0040	0.0059	(0.0141)
Profit changes		Parameter	Mean	SD	Max	90th pctl	75th pctl	Median	25th pctl	10th pctl	Min
1	1	<i>Intercept</i>	0.0060	0.0002	0.0063	0.0062	0.0061	0.0060	0.0059	0.0058	0.0054
1	1	<i>model_Δ</i>	0.0923	0.0064	0.1119	0.1008	0.0965	0.0918	0.0891	0.0838	0.0781
2	2	<i>Intercept</i>	0.0040	0.0002	0.0045	0.0042	0.0041	0.0040	0.0038	0.0037	0.0034
2	2	<i>model_Δ</i>	0.1072	0.0082	0.1284	0.1168	0.1132	0.1081	0.1014	0.0958	0.0892
2	2	<i>HRR</i>	(0.1047)	0.0070	(0.0017)	(0.0075)	(0.0097)	(0.0135)	(0.0196)	(0.0252)	(0.0329)
3	3	<i>Intercept</i>	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
3	3	<i>model_Δ</i>	0.1036	0.0111	0.1385	0.1168	0.1120	0.1039	0.0959	0.0906	0.0680
3	3	<i>HRR</i>	0.0064	0.0114	0.0377	0.0198	0.0142	0.0062	(0.0010)	(0.0086)	(0.0189)
3	3	<i>CL</i>	(0.0330)	0.0096	(0.0137)	(0.0201)	(0.0253)	(0.0331)	(0.0408)	(0.0465)	(0.0528)

(Continued)

Table 10
(Continued)
B. 5-year horizon

Simulated distributions										
Value changes	Parameter	Mean	SD	Max	90th pctl	75th pctl	Median	25th pctl	10th pctl	Min
1	<i>Intercept</i>	0.0078	0.0004	0.0087	0.0084	0.0081	0.0078	0.0075	0.0071	0.0069
1	<i>model_Δ</i>	0.0059	0.0020	0.0117	0.0082	0.0074	0.0057	0.0044	0.0034	0.0011
2	<i>Intercept</i>	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2	<i>model_Δ</i>	0.0079	0.0024	0.0144	0.0105	0.0097	0.0079	0.0065	0.0047	0.0026
2	<i>HRR</i>	(0.0333)	0.0040	(0.0231)	(0.0284)	(0.0304)	(0.0334)	(0.0359)	(0.0385)	(0.0427)
3	<i>Intercept</i>	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
3	<i>model_Δ</i>	0.0100	0.0032	0.0184	0.0139	0.0125	0.0096	0.0078	0.0063	0.0006
3	<i>HRR</i>	(0.0161)	0.0080	(0.0015)	(0.0066)	(0.0105)	(0.0151)	(0.0216)	(0.0280)	(0.0337)
3	<i>CL</i>	(0.0206)	0.0090	0.0012	(0.0087)	(0.0143)	(0.0209)	(0.0273)	(0.0308)	(0.0411)
Profit changes	Parameter	Mean	SD	Max	90th pctl	75th pctl	Median	25th pctl	10th pctl	Min
1	<i>Intercept</i>	0.0071	0.0001	0.0074	0.0072	0.0071	0.0071	0.0070	0.0069	0.0067
1	<i>model_Δ</i>	0.0802	0.0054	0.0930	0.0878	0.0842	0.0797	0.0763	0.0731	0.0698
2	<i>Intercept</i>	0.0057	0.0001	0.0063	0.0059	0.0058	0.0057	0.0057	0.0055	0.0053
2	<i>model_Δ</i>	0.0946	0.0067	0.1072	0.1040	0.0996	0.0946	0.0901	0.0863	0.0779
2	<i>HRR</i>	(0.0311)	0.0051	(0.0187)	(0.0246)	(0.0273)	(0.0312)	(0.0345)	(0.0380)	(0.0425)
3	<i>Intercept</i>	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
3	<i>model_Δ</i>	0.0816	0.0085	0.0990	0.0937	0.0882	0.0824	0.0744	0.0708	0.0646
3	<i>HRR</i>	(0.0199)	0.0069	(0.0056)	(0.0116)	(0.0153)	(0.0195)	(0.0253)	(0.0295)	(0.0378)
3	<i>CL</i>	0.0055	0.0065	0.0194	0.0138	0.0106	0.0051	0.0015	(0.0043)	(0.0087)

Notes: This table shows the distribution of parameters from regressions that are identical to those in Table 8 except that we use simulated parameter estimates to generate forecasts. The simulated values are based on the point estimates and variance-covariance matrix from the first-stage estimation of Equations (4) and (19), as summarized in Table 2. We conduct 100 simulations for each forecast horizon. The values in the table show the distribution of parameter estimates from period-ahead predictive regressions of quarterly changes in rival firm value ($\Delta value$) and profitability ($\Delta profit$) on model-implied changes ($model_{\Delta}$), as well as the predicted changes based on the explanatory variables from Hsu, Reed, and Rocholl (2010) and Chod and Lyandres (2011). The predictive regressions are estimated using out-of-sample data from quarters 12 to 24 (3-year horizon) and 20 to 40 (5-year horizon) and from quarters *HRR* and *CL* represent residuals from regressions using the *HRR* and *CL* variables as defined in Table 5. In Specification 1, *model_Δ* is the only explanatory variable. Specifications 2 and 3 add *HRR* and *CL*, respectively. * $p < .05$; ** $p < .01$; *** $p < .001$.

Table 11
Explanatory variable ranks (out-of-sample model selection)
Out-of-sample predictive regressions

Dep. var.	HRR variables				HRR and CL variables (negative CSM subsample)			
	ΔV		$\Delta profit$		ΔV		$\Delta profit$	
	3-year	5-year	3-year	5-year	3-year	5-year	3-year	5-year
<i>model_Δ</i>	1	1	1	1	1	1	1	1
<i>HRR</i>	Exclude	2	2	2	Exclude	Exclude	2	Exclude
<i>CL</i>	n/a	n/a	n/a	n/a	Exclude	2	Exclude	2

Notes: This table summarizes the ranks of each of the explanatory variables using model selection based on the Schwartz-Bayesian information criterion. The table shows results from period-ahead predictive regressions in which the candidate variables are model-implied changes in firm value and profitability, as well as the changes predicted by the variables in Hsu, Reed, and Rocholl (2010) and Chod and Lyandres (2011). These are *model_Δ*, *HRR*, and *CL*, respectively. Tables 5 and 8 define these variables. n/a indicates that the CL model is not applicable (i.e., when the sample includes both positive and negative CSM industries).

4. Structural Forecasts: An Analog to Difference-in-Differences

A dynamic structural model like the one in this paper produces forecasts that adjust for underlying attributes that may vary across firms and over time. The resultant forecasts allow tests that are analogous to a difference-in-differences (DD) type analysis, but can be used in settings in which traditional DD is difficult to implement. DD can be problematic when there are few appropriate control firms, when the event of interest causes changes to occur gradually (making event windows difficult to define), or when pre-event data are missing for some firms of interest. Beyond the analog to DD, the structural model also generates parameter estimates that can further refine the conclusions one draws from the data.

In traditional DD, an outcome variable *X* within a population *A* undergoes a treatment. Researchers compare the change in outcome variable *X* in population *A* to the change in outcome variable *X* within a control population *B*, which does not undergo the treatment. Assuming well-matched populations, the difference in the changes in *X* across populations *A* and *B* can reasonably be attributed to the treatment. DD is a powerful empirical tool; however, data limitations can restrict its use in some settings. For example, finding same-industry control firms may not be easy when there are only a handful of firms from which to choose. The size of the available pool naturally limits the number of attributes on which firms can be matched. Forced compromises then may yield inadequate matches that generate false positives in the empirical tests. This can happen when industry shifts, whether because of increasing competition from a new entrant or evolution in consumer tastes, affect some firms more intensely over time than others. A truly dynamic oligopoly model offers researchers a way to incorporate potentially important heterogeneity across firms by allowing changes in the outcome variable of interest to depend on both a company’s characteristics and the time elapsed. In this way, dynamic

models can potentially avoid false positives in cases where static models may not.

To clarify the benefits of estimating a dynamic model, consider the following analogy. Several horses are training for a series of races. A researcher develops a model that uses the horses' training and last-race times to forecast how they will perform from race-to-race. After the model is estimated, a few new horses arrive, begin the training protocol, and enter the race series. In addition to the standard training protocol, these new horses are given a special feed that, in reality, is useless. Using the model based on the data from the untreated horses, forecasts of future performance are created for all of the horses. The difference between the actual race times and the forecasts are the forecast errors. If the feed is helpful, then the forecast error for the treated horses should differ from those of the other horses in the race (i.e., they should overperform).

How does a dynamic model help reduce the likelihood of a false positive? In this example, one empirical complication might be that the new horses' times improve at faster rates. A DD model would typically forecast the time in race $n + 1$ to be equal to the time in race n . A dynamic model would consider the time in training. Improvement over time is expected, and a reasonably well-estimated model should show that it occurs at a declining rate. In this case, the DD model would generate a false positive because the feed only goes to the horses that have not been in training for very long. A dynamic model would adjust for time in training and indicate that the feed is not helpful.

5. Related Work on Intraindustry Effects

While the IPO literature is voluminous (for surveys, see Ritter and Welch 2002; Ljungqvist 2007), we are aware of only three other articles—HRR, CH, and CL—that explore empirically how a firm's decision to go public affects the values of other firms in its industry. In each case, the authors document deteriorating rival firm performance post-IPO, and they point to a strong IPO firm as the likely cause. Our results indicate that, in most cases, IPOs simply presage changes in an industry. They do not induce them. This is consistent with the fact that most firms conducting an IPO are small and stay small over the next several years.

Figure 2 reveals that market shares of more than 10% are relatively rare, comprising only 1.1% of all IPO firms using 2-digit industries and 12.2% of all IPO firms using 4-digit industries. At the other end of the spectrum, small IPO firms are rather common. In industries defined at the 2-digit level, 92% of the firms undertaking an IPO have market shares under 1.5%. Using 4-digit industries, this figure naturally drops, but still comes in at 57%. Three years

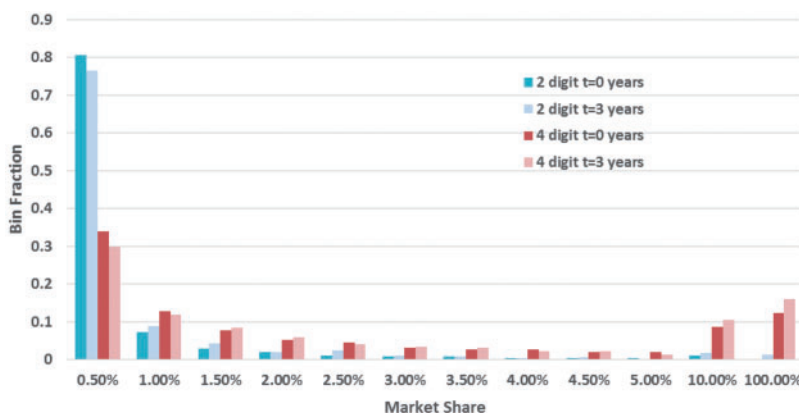


Figure 2
IPO firm market share.

Figure 2 displays the IPO firm's Compustat market share as of its IPO date and three years following the IPO. Industry is defined according to both 2- and 4-digit Standard Industrial Codes. Bin labels represent the upper bound. For example, the bars in the 0.50% bin represent firms that went public with market shares less than or equal to 0.50%.

later, the IPO firms remain small or disappear altogether.³² In the subsequent 3 years, 94% of the IPO firms see their market shares change in absolute value by less than 1% using 2-digit industries. Even at the 4-digit level, more than 60% see changes in their absolute market shares of less than 1%.³³ Consistent with our forecast error tests, this suggests that a firm's decision to go public has a rather modest practical impact on its competitive stance, limiting the firm's direct impact on rivals.³⁴

Our paper is also related to the substantial empirical literature documenting intra-industry spillover effects of firm-level announcements outside of the IPO context, including: mergers and acquisitions (Eckbo 1983, 1992; Song and Walkling 2000; Fee and Thomas 2004; and Shahrur 2005); dividend announcements (Laux, Starks, and Yoon 1998); bankruptcies (Lang and Stulz 1992); corporate security offerings (Szewczyk 1992); and cash policy (Fresard 2010). As in Spiegel and Tookes (2013a), the advantage of the model in this paper is that it produces a testable structure for examining the mechanisms driving these spillover effects.

³² Approximately one-fifth of IPO firms (20.5%) in our sample disappear from the public markets. This finding is consistent with Bharath and Dittmar (2010), who find that IPO firms often return to private ownership within a few years.

³³ Our data parallel the results in Chemmanur, He, and Nandy (2010). This rather mundane productivity performance for the IPO firms is also accompanied by long-run stock return underperformance (Ritter and Welch 2002).

³⁴ Market shares for IPO firms that we observe through year $t + 3$ do increase on average; however, we observe 3,299 IPO firms at date 0 and only 2,569 by the end of year 3 following the IPO. Although some of the firms disappear from the sample because of M&A activity, many also disappear because of failure (i.e., market shares approach zero).

Within the corporate finance literature that employs structural models, a paper analogous to ours in its focus on reexamining an existing conclusion through a structural model is Coles, Lemmon, and Meschke (2012). The authors estimate a static principle-agent model to examine the hump-shaped relationship between Tobin's q and managerial ownership. Prior papers interpreted this relationship to imply that having managers hold additional corporate shares at low levels improves their alignment with outside investors but that high levels of ownership encourage entrenchment. The authors show that their structural model of optimal contracting fits the data well and that the hump shape does not arise from a misalignment of managerial holdings and shareholder interest, as prior authors claimed. They find purely empirical models can lead to false positives that structural models can sometimes avoid.

In addition to the IPO and intra-industry spillover literatures, this paper adds to the growing body of structural corporate finance research. Prominent examples include Hennessy and Whited (2005, 2005), Strebulaev (2007), and Riddick and Whited (2009), which focus on capital structure and investment dynamics.³⁵ These papers provide tests of quantitative predictions, in addition to qualitative analyses (i.e., tests of dominant effects). Like these papers, ours is clear about the objective functions of firms and the ways in which the firms' choices over time affect future dynamics. Our contributions lie not only in the IPO application but also in the ability to characterize the value dynamics of entire industries.

6. Conclusion

Our results indicate that rival firms incur losses in the majority of IPOs, and that these losses are driven by industry trends. The IPO is just a portent of those trends. The dynamic model used here reveals that most post-IPO changes arise because profits tend to get squeezed as consumers increasingly view the industry's products as less distinct, making it easier for one firm to steal another's customers. Of course, this is not true of all IPOs. In about 7% of all cases, the dynamic model indicates that, the newly public firm does get stronger post-IPO, leading to a deterioration in its rivals' performance.

To verify that the estimated model is producing meaningful forecasts, we test it out-of-sample. We find that the model performs well and that when we add additional variables from the prior literature, the model remains the most important predictor of post-IPO changes in both profitability and value.

While it is important to document the changes in rivals' performance following an IPO, traditional empirical approaches make the drivers of these changes difficult to identify. The structural model used here offers a way to uncover the economic channels driving the changes in performance following

³⁵ See Strebulaev and Whited (2012) for a survey.

an IPO. On average, the evidence suggests that all firms suffer post-IPO. The IPO's trigger seems to be increased consumer competition. When consumers become easier to steal, private firms go public.

Appendix

A.1 Derivation of Equation (7)

Write the value function as

$$V(m, \zeta, t) = E_P \left[\int_0^\infty e^{-(rt - \zeta t)} \pi_i(t) dt \right] \quad (\text{A.1})$$

under some probability measure P . This can be changed to an equivalent measure Q using the Radon-Nikodym density $e^{-\left(\frac{1}{2}\sigma^2 t - \sigma W_t\right)}$ to transform (A1) to

$$E_P \left[\int_0^\infty e^{-(r - g - \frac{1}{2}\sigma^2)t} e^{-\left(\frac{1}{2}\sigma^2 t - \sigma W_t\right)} \pi_i(t) dt \right] = E_Q \left[\int_0^\infty e^{-(r - g - \frac{1}{2}\sigma^2)t} \pi_i(t) dt \right]. \quad (\text{A.2})$$

This creates an equivalent value function without the ζ term, effectively changing the discount rate from r to $r - .5\sigma^2$ in the HJB equation.

A.2 Endogenizing the IPO

Firms begin with private financing and with market shares that are below their long run equilibrium values. During their lives, they face a trade-off. Remaining private allows them to keep competitive secrets concealed and reduces regulatory costs. Going public reduces capital costs, and it may encourage (now better-diversified) executives/owners to take on high-risk, but ex ante, profitable projects.³⁶ Within the model, it must be that the private financing costs and benefits in terms of α , s , and f yield a lower value relative to those from going public at some future point in time. A reduction in capital cost is equivalent to assuming that $\alpha_{ip} \leq \alpha_i^*$ and/or that $s_{ip} \leq s_i^*$, where the subscript p stands for the parameter's value while the firm is private. Although every firm might like to become a stronger competitor via lower-cost financing and lower managerial risk aversion, the fixed regulatory costs of going public are not trivial. Assume it costs f to go public and that the instantaneous fixed operating costs are lower if the firm is private rather than public, $f_{ip} \leq f_i^*$ and $f_{ip} \leq f_i$. Offsetting these advantages to public financing is the fact that private firms are better able to keep their competitive advantages secret. For algebraic simplicity, assume this means the industry parameters α , s , and f remain unchanged, while the firm is private (i.e., ψ equals 0).

³⁶ This is the hypothesis tested in CL.

While firm-specific factors may induce a firm to go public, a viable alternative explanation is that an industry-wide change triggers the decision. For example, over time firms are likely to copy each other's successful product lines and production processes. Even though this increased homogenization makes it easier for firms to acquire each other's customers, it simultaneously reduces the value of doing so. To model this mathematically, assume that a Poisson process with parameter λ triggers (1) the consumer mobility parameter to increase from ϕ_λ to ϕ ; (2) the profit parameter to drop from α_λ to α^* (i.e., $\alpha_\lambda > \alpha^*$); and (3) the difference between α^* and α to shrink (using subscript λ to represent pre-shock values one has $\|\alpha_\lambda^* - \alpha_\lambda\| < \|\alpha^* - \alpha\|$).

When calculating the value function for the initially private firm, there are several cases to consider. (1) Is the firm private or public when the industry becomes more homogeneous? (2) After the industry changes, if the firm is still private, how much time will pass until it goes public? (3) Will the firm ever go public prior to the change in the industry? (4) What about after the change? The algebra needed to cover every situation is extensive. To keep the discussion manageable, we only detail a situation where the firm never goes public unless the going-public conditions are satisfied. For the IPO data we analyze, this seems like the most relevant case. That is, the firm stays private until the right conditions materialize and then goes public. This rules out cases in which the firm never goes public and where it immediately goes public at the model's start. The solutions, with some minor notational changes, also cover situations where the private firm never goes public and where it goes public even if the industry never becomes more homogenous. Empirically, these cases include those where an IPO is due to a firm's desire to become a stronger competitor and those where industry changes are the primary catalyst.

Solving for the case in which a firm is private unless the industry changes requires several steps. With the assumptions given in Section 1, the HJB equation has the same form as (7) with $\Delta\alpha_i$ and Δf_i set to zero. We first guess that the value function is affine in m_i and of the form $V_i(t) + a_{ip}(t) + b_{ip}(t)m_i$. Dropping the assumptions required to set $\hat{\alpha}_i \tilde{s}_i \hat{z}$ to the constant $\alpha_i s_i z$ for all t post-IPO, the solution to a_i is given implicitly as the solution to the ODE, where $a_i^t = da_i/dt$,

$$0 = -e^{-\psi t} \Delta f_i - f_i - \frac{b_i \phi(n-1) [b_i \tilde{s}_i \hat{z} - (n-1)]}{\left\{ 1 + (n-1) [b_i \tilde{s}_i \hat{z} - (n-1)]^{-1} \right\}^2} - \left[1 - \frac{b_i^2 s_i \phi}{\sum_{k=1}^n \frac{b_k \phi(n-1) [b_k \tilde{s}_k \hat{z} - (n-1)]}{\left\{ 1 + (n-1) [b_k \tilde{s}_k \hat{z} - (n-1)]^{-1} \right\}^2}} \right] - \delta a_i + a_i^t \quad (\text{A.3})$$

In this case, the terminal value sets the solution to a_{ip} and b_{ip} , equal to their value in (A3) and (12) at the IPO date, t^* , with an adjustment for the fixed cost

F of going public. Some algebra shows that the solution to b_{ip} is

$$b_{ip} = \frac{\alpha_{ip}}{\phi + \delta} + \left[\frac{\alpha_i - \alpha_{ip}}{\phi + \delta} + \frac{\Delta \alpha_i}{\phi + \delta + \psi} \right] e^{-(\phi + \delta)(t^* - t)}. \quad (\text{A.4})$$

Because the solution to a_i does not have a general closed-form solution, it is not possible to produce an equation like (A4) for a_{ip} without additional restrictions to the model. Instead, let a_i^* equal the value of a_i on the firm's IPO date, which is given implicitly as the solution to (A3). For the pre-IPO period, the ODE governing a_{ip} has the same form as (A3) with k_α , k_s , and ψ all set to zero. Let $z_p = 1/\sum s_{ip}\alpha_{ip}$ then the solution to this ODE can be written as

$$a_{ip} = \left[1 - e^{(t-t^*)} \right] \left\{ -\frac{f_{ip}}{\delta} + \frac{\phi \alpha_{ip} [\alpha_{ip} s_{ip} z_p - (n-1)]^2}{\delta(\phi + \delta)(\alpha_{ip} s_{ip} z_p)^2} \right\} + e^{(t-t^*)} [a_i(t^*) + F]. \quad (\text{A.5})$$

To find the IPO date differentiate $a_{ip} + b_{ip}m_i$ with respect to t^* , set the result equal to zero, and solve for t^* . (Note that the optimal time includes the changes in the firm's market share, because m_i also depends on t^* through (1).) It is easy to show that the smaller the firm is on the day the industry shock hits, the longer it will wait to conduct an IPO. The private firm becomes a stronger competitor if it goes public, so it is also straightforward to show that both db_{ip}/dt^* and da_{ip}/dt^* are negative (the latter if f_i is not too large relative to f_{ip}). Thus, the firm will only remain private if $d(b_{ip}m_i)/dt^*$ is positive. This may be true for an initially small market share. However, as m_i approaches its steady-state value, dm_i/dt goes to zero. Thus, for the appropriate parameter values, there is an optimal IPO date. Not surprisingly, increasing the regulatory costs of going public implies that firms will be larger in size when they finally conduct their IPOs. That is, $dt^*/d\alpha_{ip} > 0$ for $\alpha_{ip} > \alpha_i$.

To complete the analysis of the case in which the firm remains private unless there is a change to the industry after which it will (at some point) go public, begin by considering the value function prior to the change. To keep the solution as simple as possible, assume the private firm reaches its steady-state market share prior to the date of the industry shock. In this case, the appropriate guess for the value function is that it is time independent. This leads to an HJB equation of the form

$$0 = \max_{u_i} \alpha_{i\lambda} m_i - f_{i\lambda} - u_i + V_{i\lambda}^m \phi_\lambda \left[\frac{s_i u_i}{\sum_{j=1}^n s_j u_j} - m_i \right] - \delta [\lambda (a_{ip} + b_{ip} m_i) + (1 - \lambda) V_{i\lambda}]. \quad (\text{A.6})$$

The subscript λ indicates a parameter's value prior to the industry shock.³⁷ In (A6) the a_{ip} and b_{ip} terms equal their value immediately post shock; this occurs

³⁷ If the private firm has not yet reached its equilibrium market share then (A6) needs to be adjusted for the fact that the time between when the industry shock happens and the firm goes public depends on when the shock occurs. If the private firm has already reached its steady-state market share, the time between these two events is a constant.

at time $t = 0$ in (A4) and (A5). Next, guess that the value function in (A6) has the form $V_{i\lambda} = a_{i\lambda} + b_{i\lambda}m_i$, where $a_{i\lambda}$ and $b_{i\lambda}$ are both time independent. Quite a bit of algebra then yields solutions of

$$a_{i\lambda} = \frac{\lambda - 1}{\lambda} a_{ip} - \frac{f_{i\lambda}}{\delta\lambda} + \frac{\phi [\alpha_{i\lambda}(\phi + \delta) - \alpha_i^* \delta(1 - \lambda)]}{\delta\lambda(\phi + \delta)(\phi + \lambda\delta)} \left[\frac{(\alpha_{i\lambda}(\phi + \delta) - \alpha_i^* \delta(1 - \lambda)) s_{i\lambda} z_\lambda - (n - 1)}{(\alpha_{i\lambda}(\phi + \delta) - \alpha_i^* \delta(1 - \lambda)) s_{i\lambda} z_\lambda} \right]^2 \quad (\text{A.7})$$

and

$$b_{i\lambda} = \frac{\alpha_{i\lambda} - \delta(1 - \lambda)b_{ip}}{\phi + \lambda\delta}. \quad (\text{A.8})$$

In the above equations, $z_\lambda = 1 / \sum (\alpha_{i\lambda}(\phi + \delta) - \alpha_i^* \delta(1 - \lambda)) s_i$.

Given the above set of solutions, with sufficient pre-IPO period data, it is possible to investigate the degree to which the model does or does not explain IPO activity. While predicting IPO activity is not the main research question in this paper, a number of the comparative statics from the endogenous IPO case can help us interpret the estimates from the main model. For example, greater product homogenization (higher ϕ) leads to lower profits and simultaneously reduces the optimal size at which a firm should go public. Thus, industries where ϕ increases will see some firms go public that were previously disinclined to do so. In these cases, IPOs just presage declines in industry value rather than cause them. This is an alternative hypothesis to the one in which the IPO firm is itself responsible for the decline in rival values, and we examine it in detail in the empirical analysis described in Section 3.

A.3 Quarterly Data Alignment (Note)

Firms within industries sometimes have heterogeneous fiscal end dates. To construct industry market shares using quarterly data, one should align the reported values. To do so, we implement the following smoothing rule. Where the fiscal quarter end date does not equal the calendar quarter end date, we use data from the last report date preceding the calendar quarter end as well as the first quarter following the quarter end. Fiscal quarters are transformed to calendar quarters via weighting the consecutive fiscal quarter values by the distance to the calendar quarter end date.

A.4 Filtering Implausible Model Estimates

Table 2 indicates that an empirical model that simply says industries lose value post-IPO may be missing some important heterogeneity in the data, even if that is the median result. Of course, like any structural model, the one estimated here is clearly unable to fit some industries and firms. Table 2 shows large differences between 95th percentile and maximum values for all parameters. To reduce the impact of outliers, we eliminate IPO events with estimated ψ , k , g , or ϕ that are greater than the 99th percentile or less than the 1st percentile of all estimates. Because it is unlikely that a forecaster would use extreme or unreasonable estimates for prediction, we filter the value change data by

removing from the sample all firm-quarter observations in which model-implied or actual quarterly changes in firm value (log ratio of values) are less than -1 or greater than +1. We also remove observations in which we observe very large swings in profitability, which we define as changes in model-implied or actual quarterly changes in profitability that are greater (in absolute value) than the value of beginning-of-period assets.

Taken together, these three IPO event and firm-level filters reduce the initial sample from 128,859 firm-quarter observations to 62,497 in the 3-year sample and from 227,583, to 118,499 for the 5-year sample. These firm-quarter values include all firms in an industry, not just the IPO firm. Approximately 10% of the reduction in sample size comes from the event-level filters, with which we remove ψ , k , g , or ϕ that are greater than the 99th percentile or less than the 1st percentile of all estimates. The rest is due to removing unreasonable estimates at the firm-quarter level and is driven by small firms. In an effort to preserve as much data in the initial sample of rivals as possible, we did not impose an initial market share filter (as in Spiegel and Tookes 2013a). We think that this choice is appropriate in the IPO setting. However, it is also the case that small firms' changes in fundamentals are much noisier than those of larger firms; therefore firm-level parameter estimates for these firms are likely to be less stable. Using the 5-year estimation horizon as an example, when we compare the market shares of firms that are included in the estimation to those that are not included, the median shares are 2.7% and 0.95%, respectively.

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